Does recursion amplify capability in the same way across very different fields?

Theory · The Foundational Paper of the ARC Theory, within Recursive Dynamics: the foundation of Law I, the ARC Principle, a law under test

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Abstract

I believe advanced AI is an existential risk unless values and correction are built into how it reasons; only proof will move the world. If control of a capable AI system were lost and could not be won back, what the system itself carries would decide what it does. This paper asks whether recursion amplifies capability in the same way across very different fields, so that a self-improving system's growth could be predicted before it scales. Between 2024 and 2026, programmes in AI reasoning, quantum error correction, acoustic physics and consciousness science reported results that bear on recursive or recurrent processing, in different forms. We formalise the question with three axioms (separability, cumulative advantage, continuity) and a recursive composition operator. Under each case's hypotheses, Cauchy's functional equations give multiplicative composition a power law and additive composition an exponential; capped composition stays bounded. Inside the model the asymptotic exponent is derived, not fitted: $\alpha = 1/(1-\beta_L)$ for coupling β_L; base potential and depth must interact multiplicatively; and a crossover depth $R^* = \alpha/a$, independent of base potential, separates approximately linear from compounding gains. Five structural properties follow, two on premises added to the axioms. The ARC Bound is stated as registered: the ceiling of Law III, the ARC Ceiling, a law under test, is at most two for same-class correctors, and the point value two is a conjecture. The evidence is graded: the AI exponent's one interpretable fit is inconclusive, the time crystal is an analogy with no exponent measured, and the neuroscience mapping is structural. Thirteen falsification criteria and a cross-domain forward prediction are stated; a computational suite checks the closed forms but tests no real system. Each result holds inside its stated model.

Version 5.21 states Unit 83's registered result, states the ARC Bound as the theory registration states it, grades the runs of March 2026, the alignment run and the time crystal as their sources and the corrections register now do, and moves the dated notes the body kept beside corrected text into Appendix B, which lists every version and correction, with its date.

The Foundational Paper of the ARC Theory, within Recursive Dynamics: the cross-domain foundation of Law I, the ARC Principle, and the formalism of the growth exponent that Law III, the ARC Ceiling, caps, each a law under test; it states the ARC Bound as the theory registration holds it, a ceiling of at most two for correctors of the system's own class, whose point value two is a conjecture, and it bears on Law V, the ARC Embedding Law (added on 24 September 2026 and registered as a law on 15 October 2026 in version 1.103 of that registration, a law under test, the registered result on its placement limb given in Section 8; its mathematics in development), through the conditional prediction of Section 6.5, beside which it reports a pilot of the Eden Protocol. Paper I states Law I formally. Working paper, version 5.21, 31 October 2026; first published 13 February 2026; Appendix B lists every version and correction. Recorded on OSF as *Foundational Paper · The ARC Theory · Law I, the ARC Principle · Recursive Amplification as a Cross-Domain Structural Principle*, DOI 10.17605/OSF.IO/Y7QGD. About the author, founder of Recursive Dynamics, without an institution or a doctorate: michaeldariuseastwood.com/questions/who-is-michael-darius-eastwood; his book, *Infinite Architects*, first edition as printed: osf.io/dqm3e. Licence: CC BY 4.0.

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1 Introduction

I believe advanced AI is an existential risk unless values and correction are built into how it reasons; only proof will move the world. If control of a capable AI system were lost and could not be won back, what the system itself carries would decide what it does. The programme's position is that people keep oversight, containment and shutdown for as long as they physically can, and never give them up by choice; correction built into a system is the contingency for control lost against our will, not a reason to let go of it. If the way a recursive process combines its gains fixes how its capability grows with depth, then the scaling of a self-improving system could be predicted from a property measured before it scales: if you can measure the composition operator in a new system, you can predict its scaling function before measuring the full scaling curve (Section 3.2). On that reading, structural alignment (embedding values within the composition operator itself; the values are for international consensus, Section 6.5) is to be built while systems remain frozen and their scaling is predictable, and judged on whether it keeps a system useful as well as safe. This paper asks whether recursion amplifies capability in the same way across very different fields: is the pattern that separate research programmes have reported (Section 1.1) a coincidence, or are we observing different expressions of a single structural principle?

This paper belongs to the ARC Theory, the founding theory of Recursive Dynamics, the science of systems that improve themselves, which the author founded (the founding paper). The theory is carried by five laws, each a law under test: Law I, the ARC Principle, that usable capability is a power of recursive depth, with the exponent measured rather than assumed; Law II, the ARC Co-Scaling Law; Law III, the ARC Ceiling; and Laws IV and V, the ARC Persistence Law and the ARC Embedding Law, added on 24 September 2026, their mathematics in development. The laws and twenty-two propositions are registered on OSF (the theory registration), and the statement paper states the theory whole. The programme's engineering hypothesis is the Eden Protocol, correction built into a system's own reasoning, whose pilot Section 6.5 reports; its end goal is to learn, by registered tests, whether correction built into a self-improving system can keep pace with its capability, as the contingency for control lost against our will. This paper brings four fields together, AI reasoning, quantum error correction, acoustic physics and consciousness science, and assumes no reader knows the others: Section 1.1 says what each reported, and Appendix G places the paper among the laws.

We argue that the pattern reflects a principle, but not the one that might be expected. We do not claim that one equation fits all domains. We claim something more specific: that, for any recursive system that satisfies the axioms, a measurable property (its *composition operator*) determines its scaling function, and that five qualitative properties follow from the recursive architecture, two of them on premises added to the axioms (Theorem 4). The mathematical form differs across domains because the composition operator differs. The structural properties hold in every substrate whose recursion satisfies the axioms, because the axioms name no substrate.

Our contributions are:

- **A formalism.** Three axioms, separability, cumulative advantage and continuity (Section 3.1), and the recursive composition operator $\oplus$, which characterises how incremental gains combine with accumulated capability (Definition 1), giving the generalised ARC Equation (2).
- **A classification.** Under each case's hypotheses, multiplicative $\oplus$ yields power-law scaling (R^{α}), additive $\oplus$ yields exponential scaling ($e^{\alpha r}$ in the added passes $r = R - 1$), and bounded $\oplus$ yields saturation (Theorem 1), each form stable within its own functional equation (Section 3.2).
- **The exponent derived, not fitted.** Within the stated model, the asymptotic exponent is $\alpha = 1/(1 - \beta_L)$ (Theorem 2); no additive decomposition of capability is consistent with the axioms (Theorem 2a); and the crossover depth $R^* = \alpha/a$ is independent of base potential (Theorem 3).
- **Five structural properties**, threshold behaviour, recursive depth dependence, base quality de-

pendence, multiplicative $I \times R$ interaction and regime boundaries, whose conjunction is the framework's distinctive prediction (Theorem 4), and composition operator transitions within a single bounded system (Conjecture 5, supported in three computational systems and untested in physical experiments, not derivable from the three axioms).

- **The ARC Bound, stated as registered.** The claim that the ceiling of Law III, the ARC Ceiling, is at most two for correctors of the system's own class, which needs P6 and P20 and nothing else, with the point value two a conjecture; this paper's own earlier form of it, the coupling at which the sustained exponent reaches two ($\beta_L = 0.5$, $\alpha = 2$), is an early-framework conjecture that decides nothing registered (Section 6.4).
- **Graded evidence and tests.** Evidence from four domains (Section 4), thirteen falsification criteria (Section 7), a cross-domain forward prediction (Section 6.1), a measurement protocol (Appendix E) and a computational validation suite checking the core mathematical relationships to machine precision (Appendix F).

The **ARC Principle** is Law I as the theory registration states it: usable capability is a power of recursive depth, with the exponent measured rather than assumed. The first four items are this paper's framework for it, and the conjunction of the five properties of Theorem 4 is the framework's own claim, tested by criterion F4; Law I takes its word through P1 alone, so no result on F4 refutes Law I. The power law is the form the framework derives for multiplicative composition, and the registration places the cross-domain claims beside Law I, as P9 and P18. The fifth item belongs to Law III, the ARC Ceiling, a law under test.

Section 1.1 sets out the four programmes and the question they raise, and Section 1.2 the ARC Equation and its generalisation. Section 2 places the work among related work, antecedents and the rival instrument. Section 3 gives the formalism: the axioms, the composition operator and Theorems 1 to 4. Section 4 reviews the evidence property by property, Section 5 explains the forms across domains, and Section 6 states the predictions and the ARC Bound as registered. Section 7 lists the falsification criteria, Section 8 says what the paper does and does not show, Section 9 states its limitations, and Section 10 concludes. Appendix A gives the proofs, Appendix B the version history and every correction, Appendix C the priority record, Appendix D the notation, Appendix E the measurement protocol, Appendix F the computational validation suite, and Appendix G the paper's place in the ARC Theory.

1.1 The problem

Between August 2024 and February 2026, four research programmes reported the following results:

Google Quantum AI (results announced 9 December 2024; the team's preprint public from 24 August 2024) demonstrated exponential suppression of the logical error rate below the surface-code threshold on the Willow processor, by a factor $\Lambda = 2.14 \pm 0.02$ for each increase of two in code distance (Google Quantum AI and Collaborators 2024), a result graded here as convergent timing, not a prediction.

DeepSeek-AI (preprint January 2025; in *Nature*, September 2025) showed that "the reasoning abilities of LLMs can be incentivized through pure reinforcement learning (RL)", with the emergence of reasoning patterns such as self-reflection and verification (Guo et al. 2025). Sharma and Chopra (2025) reported that sequential refinement, in which chains build on previous attempts, outperformed parallel self-consistency at matched compute in 95.6% of configurations, across five open-source models and three reasoning benchmarks. Their paper is dated before the experiment reported here, so it is the earlier demonstration of that advantage and not a confirmation of this work. The direction is also not settled: Gu et al. (2026), posted in April 2026, report the opposite result, parallel sampling outperforming sequential across three model families on mathematics and coding, attributing it to reduced exploration when a chain conditions on its own previous answers.

NYU Physics (February 2026) created a continuous classical time crystal with a pair of acoustically

levitated beads: "Unless the particles have identical scattering properties, their wave-mediated interactions are nonreciprocal", and pairs can use this nonreciprocity "to harvest energy from the wave to sustain steady-state oscillations despite viscous drag and the absence of periodic driving" (Morrell et al. 2026). The oscillation is a linearly stable limit cycle. In one experiment the antisymmetric mode's coherence time is $\tau = (100 \pm 10)$ s, about 6,700 cycles at 66.787 Hz, comparable to the 108 s recording, and the authors add: "Sustained oscillations persist for hours longer than the measurement window, which means that the true coherence time of the nonequilibrium steady state may be substantially larger." Every recording the paper reports is of a steady state, and no growth exponent has been measured in this system.

The COGITATE Consortium (2025) found information about conscious content in visual, ventrotemporal and inferior frontal cortex, content-specific decoding in posterior regions generalising for about the duration of the stimulus (0.5 to 1.5 s) while in prefrontal regions its temporal generalisation was restricted to about 0.2 to 0.4 s (Cogitate Consortium et al. 2025). COGITATE tested two theories of consciousness, not a recursion depth, and its results challenge key tenets of both; this paper reads the sustained, content-specific posterior responses as structurally consistent with depth-dependent, feedback-driven processing, which COGITATE did not measure.

These programmes address different problems in different physical substrates. They report different things: exponential suppression of error in quantum error correction, a contested sequential advantage in AI reasoning, a sustained steady-state limit cycle in a classical time crystal, and no measured form in neuroscience. They are separate programmes in separate fields.

Yet, on this paper's reading, they share a structural pattern: in each, recursive or recurrent processing (where the output of one cycle becomes the input for the next), operating on structured asymmetry, is what the reported gain or order depends on. Those programmes are cited for convergence and not for mechanism: whether any of their systems accumulates, meaning retains something from each pass whose contribution grows with depth, rather than merely conditioning on the previous pass, is untested in each case. The question this paper addresses is whether that pattern reflects a single underlying principle or mere coincidence.

1.2 The ARC Equation

We formalise the question as follows. For AI systems, the equation is

$$U = I \times R^{\alpha}, \quad \alpha \leq 2, \tag{1}$$

the ARC Equation (AI power-law form), with the ARC Bound, the registration's claim that the ceiling of Law III, the ARC Ceiling, a law under test, is at most two for correctors of the system's own class; Paper X supersedes this fixed-exponent framing as the operative safety criterion and re-scopes, without retracting, the equation and the bound (Section 6.4). Here α is the measured exponent (on today's frozen models, one model's fit and inconclusive; Section 4.3 prints its row whole). The two is not a property of the equation: it is the value Law III's ceiling, $1/(1-\gamma)$, returns where the correction-leverage exponent γ is one half, a point value the registration conjectures and does not assert (Section 6.4). Law II, the ARC Co-Scaling Law, a law under test, is stated as the registration states it, $\beta_C > k$: correction has the asymptotic scaling advantage where its exponent exceeds the drift exponent, burden has it where the correction exponent is the smaller, and at equality coefficients, delay, backlog and saturation decide the outcome. In Paper X's minimal model, with positive gain drift, $\beta_C > k$ decides whether the misalignment fraction vanishes, and with level drift as well the condition is $\beta_C > \max(k, 0)$.

Here U denotes effective capability (the letter's earlier senses in this programme are recorded in Appendix D.1), I denotes base potential (structured asymmetry; the notation register's base intelligence, what the system could do before any self-improvement), R denotes recursive depth, and α denotes the scaling exponent. **Within the stated model this paper derives α rather than fitting it**: the asymptotic

exponent follows from the self-referential coupling β_L via $\alpha = 1/(1-\beta_L)$. That the exponent is bounded by two is **not a result of this model**: it belongs to Law III, and Section 6.4 states it as the registration does, a claim on the correction-leverage exponent and not on the coupling. In the v5 cross-architecture experiment of Paper II, the one interpretable sequential fit, on Gemini 3 Flash, is inconclusive, with both point estimates below one (Section 4.3); the sequential advantage was observed only where both exponents could be estimated, and super-linear scaling ($\alpha > 1$) was not. The original prediction of super-linear scaling for current models was not borne out (Section 5.1); the mathematics is unaffected, since it does not fix β_L, but the empirical AI scaling regime is weaker than originally predicted.

But this power-law form is itself a special case. Define the **recursive composition operator** $\oplus$, which characterises how incremental gains combine with accumulated capability. Under continuity and associativity, $\oplus$ determines the system's scaling function uniquely:

$$U(R) = I \times f_{\oplus}(R, \beta_L), \tag{2}$$

the generalised ARC Equation. Under the regularity and composition hypotheses of Theorem 1: multiplicative $\oplus$ yields power-law scaling (R^{α}), additive $\oplus$ yields exponential scaling ($e^{\alpha r}$ in the added passes $r = R - 1$), and bounded $\oplus$ yields saturation.

2 Related work, antecedents and the rival instrument

The formalism rests on mathematics far older than this programme, and much of its cross-domain evidence is other researchers' work, beside the programme's own experiment, comparisons and pilot (Sections 4.3, 6.1 and 6.5); this section states each debt in the source's own terms and then weighs the nearest work.

Functional equations and their stability. The classification of Theorem 1 rests on Cauchy's functional equations (Cauchy 1821); that a function approximately satisfying one of them lies near an exact solution of the same equation is the stability theorem of Hyers and Ulam (Hyers 1941; Ulam 1960). The renormalisation-group parallel of Section 3.2 draws on Kadanoff's scaling picture for Ising models near the critical point (Kadanoff 1966) and on Wilson's renormalisation group (Wilson 1971).

The dimensional formula. Section 6.4 sets out the earlier derivations of $\alpha = d/(d+1)$, a formula this paper does not claim (West et al. 1997, 2004; Banavar et al. 1999, 2010; Demetrius 2008; Demetrius and Tuszynski 2010; Zhao 2022; Bettencourt 2013); what it adds is an account of why those derivations converge.

The forms, conceded. The relation $\alpha = 1/(1-\beta_L)$ of Theorem 2 is the exponent of the elementary solution of the separable equation Axiom 2 states. The same structure, a stock that grows at a rate proportional to a power below one of itself, is the knowledge production function of Jones (1995), whose long-run growth rate carries the factor $1/(1-\phi)$; the expression $1/(1-\beta_L)$ is also the sum of a geometric series, familiar as the Keynes multiplier (Keynes 1936) and as Dyson resummation (Dyson 1949), as Paper X's novelty ledger records of this paper. Axiom 2's name, cumulative advantage, is the term of Price (1976). What this paper adds is not these forms but the axioms that select among them and their reading across domains.

The four programmes and their neighbours. The programmes of Section 1.1 are cited for convergence, not mechanism. A photonic metamaterial analogue of a continuous time crystal (Liu et al. 2023) and a continuous space-time crystal state driven by nonreciprocal optical forces (Raskatla et al. 2024) have

also been reported; recurrent processing as a basis of consciousness has its own literature (Lamme 2006; Zheng et al. 2025); and the scaling of test-time compute in language models has been studied directly (Snell et al. 2024; Zeng et al. 2025; Li et al. 2025).

The pipes premise, with its contest. The premise that transport networks set the pace of biological growth is not this programme's claim to defend; it is the premise of a leading account in quantitative biology, and a contested one. West, Brown and Enquist derived the allometric scaling laws of biology from the geometry of nutrient-distribution networks (West et al. 1997; one of the most cited papers in its field, with an OpenAlex citation count of 5,046 as of 12 August 2026, a single-source and index-dependent count). The exact exponent remains contested: White and Seymour report basal metabolic rate proportional to body mass to the two-thirds power rather than three-quarters (White and Seymour 2003), and Kozłowski and Konarzewski have twice asked whether the model is "mathematically correct and biologically relevant" (Kozłowski and Konarzewski 2004, 2005). This programme's premise needs only the mechanism, that the delivery network sets the ceiling, and not any particular exponent; the dispute over the exponent's value leaves it untouched, but the critics question the mechanism too, so the programme takes it as a premise, and its biological reading stands or falls with it.

The exception West's own group found. The escape from the pipe limit is also not this programme's assertion; it is measured in the urban scaling literature by the same senior author. Bettencourt, Lobo, Helbing, Kühnert, and West (2007) report, in the paper's own words (their β is a city-scaling exponent, not this paper's coupling β_L): "Quantities reflecting wealth creation and innovation have $\beta \approx 1.2 > 1$ (increasing returns), whereas those accounting for infrastructure display $\beta \approx 0.8 < 1$ (economies of scale)." Infrastructure, the literal pipes, scales sublinearly; information-mediated activity scales superlinearly. They state the contrast with biology directly: "we discuss how cities are similar to, and differ from, biological organisms, for which $\beta < 1$". They derive "growth equations, which quantify the dramatic difference between growth fueled by innovation versus that driven by economies of scale" (spelling as in the original). And their closing inference names the consequence: "This difference suggests that, as population grows, major innovation cycles must be generated at a continually accelerating rate to sustain growth and avoid stagnation or collapse." On this paper's reading, that is the shape of the open problem this framework addresses: in the findings of West's group, information-mediated growth escapes the throughput limit, and the only brake in their framework is external and must be applied again and again, faster and faster. No internal limit is derived. The replacement limit, a ceiling belonging to the growing system itself, is the gap.

The pipes argument at fixed throughput. Everything that grows is fed through a channel, and for everything before software, fixing the channel fixes the growth. Starve a tumour's blood supply and its growth is held back. Fix a chain reaction's fuel and geometry and it stops. Exhaust an epidemic's susceptible hosts and it burns out. Astrophysics has a named, quantitative constraint of the same kind: the Eddington limit, a luminosity ceiling where radiation pressure balances gravity, not a hard ceiling on accretion, since ultraluminous X-ray sources sit well above it (Bachetti et al. 2014). The claim here is a generalisation of limits physics already accepts, not a new species of assertion. Two hard cases are handled deliberately. Cosmic inflation is excluded by scope: it is expansion of space, not growth of a structure on a substrate, and it ended by field dynamics rather than by running out of anything. Evolution is the sharpest case and it fits: biological complexity increased for billions of years at roughly fixed solar throughput, which is growth on information at fixed pipes, and it is also glacial, and it has no internal corrector at all, since selection is external. On this framework, evolution would carry a measurable exponent far below the ceiling; that is a hypothesis about biology falling out of a framework built for software, stated without

a criterion or a planned test, and it lies outside the theory registration's scope, which places biological systems among those "for which the framework offers an analogy and no measurement". The physicist's form of the objection is Landauer's: computation is physically implemented, so software has pipes too. The fixed-substrate clock is the answer and appears beside the claim: the regime under study holds the physical substrate fixed and asks what still grows, which is why exogenous speed-ups are excluded by definition. The functional form of capability growth in recursive depth remains an empirical question, and the registered analysis does not assume the family: under P1 the power law is compared with its registered rivals, "the logarithmic form, the exponential form, a saturating form, a broken or piecewise power law with one change of regime, and the shifted finite-window solution of this framework's own growth equation", and a declining outcome is admissible.

The rival instrument. The nearest work to this programme's question, and the right paper to weigh it against, is Engels et al. (2025), first posted 25 April 2025. It asks how oversight itself scales and answers quantitatively: oversight success is modelled as a game between capability-mismatched players whose oversight-specific Elo is a piecewise-linear function of general intelligence with two plateaus, and optimal numbers of oversight levels are derived for Nested Scalable Oversight, in which trusted models oversee stronger untrusted models that then become the trusted models at the next step. The instrument is the difference. Their variable is the capability gap between overseer and overseen, measured in Elo. This programme's variable is the composition class of the corrector, measured through the corrector's own scaling exponent. They also state the design condition in slope form: oversight improves with capability when the overseer's game-specific Elo rises faster with general Elo than the overseen system's ($m_G > m_H$), and they recommend iterating an oversight design until it does. Their framework has no term for what the overseer is made of, and this paper's formalism has none either. The programme's separate prediction, outside this paper's model, is that iterated same-class oversight, of which Nested Scalable Oversight is an instance, stays capped however many rungs are added, while a cross-class corrector is not held to that cap. The theory registration's P6 states the cap for same-class correctors, at or below one half, as a conjecture on the correction-leverage exponent of Law III (the ARC Ceiling, a law under test), an exponent the registration keeps distinct from Law II's β_C. That is where the two programmes could disagree about a measurable quantity, and this paper does not decide it.

Antecedents and near-misses, each with its differential. On the question: Hutter asked directly whether intelligence can explode (Hutter 2012, arXiv, 28 February 2012), undertaking to separate speed from intelligence explosion and to consider possible bounds on intelligence, augmenting the analysis of Chalmers (2010); the question itself goes back to Good (1966). What that literature does not contain is a derived ceiling with an architecture dependence; growth exponents of the form used here are older, as this section has conceded. On impossibility: three 2025 arXiv papers argue for limits on alignment or its control (Yao 2025b, public by 13 June 2025, attested by arXiv and independently observed by the Internet Archive; Yao 2025a, 3 July 2025; and Ball et al. 2025, 9 July 2025). All three are worst-case and qualitative: measure zero, coNP-completeness, cryptographic hardness. Their measured object is verification of a fixed system, and it is closed within its own terms; this programme's measured object is a different one, persistence of alignment as capability compounds under recursive amplification, and closure of the first does not settle the second. None reports an average-case scaling exponent or a rate. The third concludes that "safety cannot be achieved by designing filters external to the LLM internals (architecture and weights)" and argues that "an aligned AI system's intelligence cannot be separated from its judgment"; that argument from filtering intractability is consistent with this programme's architecture dependence, though it does not test it. On the mechanism: that anti-correlated estimates average better than independent ones is textbook variance reduction (antithetic variates); the mechanism is not the claim.

The claim is that architecture determines whether anti-correlation is available at all, and that this caps a safety-relevant exponent. The nearest structural analogue is the quantum error-correction threshold theorem (Aharonov and Ben-Or 2008), which also converts a qualitative worry into a critical value; it concerns physical error rates in a fixed architecture, not a corrector's scaling exponent, so it is a near-miss rather than an occupant, and the analogy was identified by this programme's own search rather than by a referee. On feedback and stability, the general proposition that inadequate corrective gain relative to system gain causes instability has a long control-theory lineage (small-gain theorems; Zames 1966); those are gain conditions on interconnected systems, not a power-law exponent criterion on a corrector's scaling with the capability of a recursively improving system. The nearest quantitative neighbour is Liu and Meng (2026), which recasts self-correction as a closed-loop control problem via a two-state Markov model and derives a directly measurable stability threshold, iterate only when the error-correction rate over the error-introduction rate exceeds $\mathrm{Acc}/(1-\mathrm{Acc})$. The differential: theirs is a per-step rate threshold at a fixed capability level, deciding whether another iteration helps now; this programme's criterion is a scaling relation across capability, deciding whether corrective strength keeps pace as the system improves. Theirs has no scaling exponent on the corrector and no capability-growth exponent, the two quantities the programme's co-scaling criterion compares; the two models meet where growth stops, as Paper X shows. Smolin's cosmological natural selection (Smolin 1992) is the antecedent for selection-shaped universes, and the programme's own December-era notes cite it contemporaneously.

A programme hypothesis beyond the model. Oversight arrangements that place a human in the loop, as amplification and reinforcement learning from human feedback do, are cross-class by construction, because the human corrector does not share the model's substrate. The programme therefore hypothesises, beyond what this paper's equations decide, that human-in-the-loop oversight shows a higher correction exponent than pure-model oversight, for a structural reason rather than a sentimental one. This is testable on existing data and does not require new systems.

3 The formalism

3.1 Three axioms

We begin with three axioms that define the recursive amplification systems this paper treats.

Axiom 1 (Separability)**.** Effective capability decomposes as $U = I \times g(R)$, where $I > 0$ is base potential (structured asymmetry; the notation register's base intelligence), $R \geq 1$ is recursive depth, and $g(1) = 1$ (at depth 1, capability equals base potential).

This axiom asserts that base quality and recursive depth contribute to capability through independent, multiplicative channels. A system with zero base potential ($I = 0$) achieves zero capability regardless of recursive depth: recursion has nothing to amplify.

Axiom 2 (Cumulative advantage)**.** The amplification factor grows according to accumulated amplification: $\frac{dg}{dr} = a \cdot g^{\beta_L}$, where the variable $r = R - 1$ tallies the passes taken after the first, so that $g = 1$ at $r = 0$ restates Axiom 1's $g(1) = 1$ in the added-recursion notation the proofs work in, $a > 0$ is a rate constant and $\beta_L \in [0, 1)$ is the self-referential coupling parameter.

This axiom captures the essential feature of recursive processing: each step's marginal amplification depends on accumulated amplification. When $\beta_L = 0$, steps are independent and amplification grows linearly. When $\beta_L > 0$, accumulated amplification accelerates future gains (super-linear scaling). The parameter β_L measures how much of accumulated context each new step can effectively leverage.

Operating on the amplification factor g rather than on absolute capability $Q = I \cdot g$ preserves the separability asserted in Axiom 1: the amplification function depends only on recursive depth, not on base potential.

Axiom 3 (Continuity)**.** The scaling function g is continuous and monotonically non-decreasing in R.

This is a regularity condition. It excludes pathological solutions and ensures measurability.

3.2 The composition operator

How does the gain from recursive step r combine with accumulated capability Q_r? We formalise this as follows.

Definition 1 (Recursive composition operator)**.** Let $\oplus : \mathbb{R}^+ \times \mathbb{R}^+ \to \mathbb{R}^+$ be the **recursive composition operator**, defined by $Q_{r+1} = Q_r \oplus \delta Q_r$, where δQ_r is the incremental gain at step r. The operator $\oplus$ characterises how incremental gains compose with accumulated capability.

The composition operator is the central mathematical object of this framework. It is what varies across domains and what determines the scaling function. It is also empirically measurable: observe how marginal gains at step r relate to accumulated capability Q_r.

Theorem 1 (Classification of scaling functions)**.** *The algebraic properties of $\oplus$ uniquely determine the scaling function $f_\oplus(R)$, under the premises each case names:*

1. ***Multiplicative composition*** *($g(R_1 \cdot R_2) = g(R_1) \cdot g(R_2)$): under Axioms 1 and 3, the unique continuous solution is $g(R) = R^\alpha$* (power law).
2. ***Additive composition*** *($f(r_1 + r_2) = f(r_1) \cdot f(r_2)$ in the added passes $r = R - 1$, with $f(0) = 1$): under Axioms 1 and 3, the unique continuous solution is $f(r) = e^{\alpha r}$, so that $g(R) = e^{\alpha(R-1)}$ and $g(1) = 1$* (exponential).
3. ***Bounded composition*** *($Q_{r+1} = \min(Q_r + \delta Q_r,\ Q_{\max})$), on the capped recurrence alone: $f(R)$ is bounded by $Q_{\max}/I$ and converges to a limit no greater than $Q_{\max}/I$, which it reaches only if the increments sum to at least the remaining gap, equality sufficing for attainment in the limit* (saturation)*. Its ceiling is separable from base potential only if $Q_{\max}$ is itself proportional to I.*

Axiom 2 is not among the premises of cases (1) and (2).

Proof sketch. Cases (1) and (2) follow from the Cauchy functional equations. For (1): let $h(x) = \ln g(e^x)$; then $h(x + y) = h(x) + h(y)$. Under continuity, $h(x) = \alpha x$, giving $g(R) = R^\alpha$. For (2): $\ln f$ satisfies $\ln f(r_1 + r_2) = \ln f(r_1) + \ln f(r_2)$, a Cauchy additive equation with solution $\ln f(r) = \alpha r$. Case (3) follows directly from the bound. Full proofs in Appendix A. □

What the classification settles is the behaviour of the three composition rules set out above. Two further continuous solutions of Cauchy's equations remain, the linear form ($f(R_1 + R_2) = f(R_1) + f(R_2)$) and the logarithmic form ($g(R_1 \cdot R_2) = g(R_1) + g(R_2)$); neither compounds, and it is these that the parallel-voting data come close to, the logarithmic one being fitted outright in Appendix E. The premises of cases (1) and (2) are functional equations on the scaling function, and case (3)'s is a capped recurrence; that a given $\oplus$ induces one of them is an assumption about the system, and the conclusions of cases (1) and (2) are Cauchy's classical results (Cauchy 1821). The composition operator therefore classifies the scaling of every recursion whose scaling function satisfies one of Cauchy's four equations or the capped recurrence of case (3); it is not a classifier of every recursion.

Frozen models versus recursive self-modification: two distinct regimes. A critical application of this classification concerns the distinction between current AI systems and hypothetical self-modifying ones. Current frontier AI models are **frozen during inference**: when they generate extended reasoning chains, the underlying weights, and with them the composition rule, remain fixed (attention patterns still vary with the input). The composition operator $\oplus$ does not change between steps. This architectural constraint confines such systems to multiplicative composition through a finite-dimensional parameter space, producing power-law fits with exponents below one, which Paper II names the physical regime, $\alpha < 1$ (Section 5.1). Empirically, Paper II's estimate of the exponent rests on one model, Gemini 3 Flash, and is inconclusive, with both sequential point estimates below one (Section 4.3 prints its row whole). Such a fit does not by itself place the model outside Axiom 2: $\alpha = 1/(1 - \beta_L)$ is the asymptotic exponent of the axiom's solution, and over a finite window, below the crossover depth of Theorem 3, that solution can return a fitted exponent below one for a positive coupling.

The Cauchy classification reveals precisely why. In a frozen system the same fixed operator composes every reasoning step, and with $\alpha < 1$ the resulting gains are real but diminishing. By contrast, a system capable of recursive self-modification, one that rewrites its own composition function $\oplus$ during operation, would occupy a qualitatively different position in the classification. If step r improves the operator used at step $r + 1$, the self-referential coupling β_L would, on this model, be positive and could rise, yielding $\alpha = 1/(1 - \beta_L) > 1$. This transition does not require quantum hardware or additive composition; it requires only that the system can modify its own reasoning architecture during inference.

No current AI system achieves this. Earlier versions of this paper did predict super-linear scaling ($\alpha > 1$) for current, frozen models (the original ARC Bound prediction, "testable within current model generations", now kept in Appendix B.5), and that prediction was not borne out; the estimate of about 2.2 is retracted (Section 4.3). On the present reading, super-linear scaling belongs to a regime that does not yet exist. The practical consequence, previewed in Section 1, is that structural alignment is to be built while systems remain frozen and their scaling is predictable, and judged beside a measure of usefulness, as every safety measure is (Section 6.5). Once self-modification arrives, the composition operator becomes a moving target, and on this model a fixed external safeguard would fall behind a capability that compounds. Correction built into the operator is the contingency for control lost against our will; oversight, containment and shutdown stay in human hands for as long as they can be held. Guaranteed controllability after self-modification is not an established result of this programme.

Claim-status ladder. **Established mathematics:** Cauchy's equations constrain recursive scaling forms. **Empirically grounded:** current frontier models are frozen during inference; the Eden pilot found stakeholder-care scores, as its rubric scored them, raised by a prompt-level intervention in all five model runs that could be analysed, under paired tests, a pilot awaiting an independent, preregistered replication. The alignment run of March 2026 is not on this rung: it cannot decide whether alignment scales with depth (Section 6.5).

Strong theoretical inference: a genuinely self-modifying system that rewrites its own composition operator during operation could enter an $\alpha > 1$ regime, and alignment would then need to be structurally load-bearing before that transition.

Conjectural extrapolation: set out at the close of the paper (Section 10).

The six hidden assumptions. Every classification theorem rests on assumptions. Theorem 1 assumes six things. Each assumption, when violated, produces a specific physical refinement rather than a contradiction:

Assumption 3 is particularly significant for the framework's applicability. The Hyers-Ulam stability theorem (Hyers 1941; Ulam 1960) proves that any function *approximately* satisfying one of Cauchy's

Table 1. The six hidden assumptions of Theorem 1.

#	Assumption	When violated	Physical example
1	Continuity	Log-periodic oscillations	Discrete lattice systems, fractal networks
2	Real-valued	Interference and phase	Quantum amplitudes, wave mechanics
3	Exact equation	Approximate solutions cluster near exact	Hyers-Ulam stability (Hyers 1941; Ulam 1960)
4	Single operator	Crossover scaling, regime transitions	Phase transitions (cf. Conjecture 5)
5	Scalar composition	Logarithmic corrections	Upper critical dimension
6	Time-independence	Dynamic traversal of solution space	Adaptive systems, machine learning

functional equations must lie close to an exact solution of *that same* equation. Formally, for the additive equation, if $|f(x+y) - f(x) - f(y)| \leq \varepsilon$ for all x, y, there is an exact additive g with $|f(x) - g(x)| \leq \varepsilon$ (Hyers 1941); for the exponential equation, and so for the multiplicative one by a change of variables, the result is superstability: a real function on a vector space with $|f(x+y) - f(x)f(y)| \leq \varepsilon$ for all x, y is either bounded or an exact solution (Baker et al. 1979). Real systems never satisfy the axioms exactly: biological networks are noisy, AI reasoning chains have variable quality, quantum error correction operates with imperfect gates. Within each equation, the corresponding form is therefore **stable** under perturbation. The earlier phrasing that made the three forms "stable attractors in function space" is withdrawn, because a function that satisfies none of the equations is not drawn toward any of them; the stability is per-equation, not a partition of function space. Whether power laws, exponentials, or logarithmic forms are empirically ubiquitous in nature is a separate question about how often systems approximately satisfy one of these equations in the first place, and Hyers-Ulam does not by itself settle it.

Assumption 4 is addressed directly by Conjecture 5 (Section 5.5): bounded systems can transition between composition operators as recursive depth increases, which could explain why different studies of the same system can observe different scaling regimes.

Structural implication: per-equation stability. Combining Theorem 1 (Cauchy 1821), the Hyers-Ulam stability theorem (Hyers 1941), and Theorem 2 ($\alpha = 1/(1-\beta_L)$): once a system is known to approximately satisfy one of Cauchy's equations, its scaling form is not a free parameter to be fitted or evolved. It is mathematically compelled by the composition operator, and within the power-law form the exponent is compelled by the self-referential coupling β_L. Each of the recognised scaling forms is *per-equation* stable: a small perturbation of an exact solution stays near an exact solution of the same equation (Hyers 1941; Baker et al. 1979). The distinction matters: 'Why is this system's exponent 2?' has two kinds of answer. The mechanistic answer describes the coupling. The mathematical answer says: no other value is possible for this composition operator with this coupling strength.

This has a crucial implication: if you can measure $\oplus$ in a new system, you can predict its scaling function *before measuring the full scaling curve*. This is the framework's strongest cross-domain prediction (Section 6.1).

Critical distinction: Cauchy constrains the form, not the exponent. A persistent misreading of the Cauchy classification is that it predicts a ceiling on recursive scaling. It does not. Cauchy's functional equations determine that multiplicative composition *must* produce power-law scaling ($g(R) = R^{\alpha}$). They say nothing about the value of α. That value is set by the Bernoulli ODE ($\alpha = 1/(1-\beta_L)$), and as $\beta_L \to 1$, $\alpha \to \infty$. There is no upper bound internal to the mathematics.

The following table makes the relationship explicit:

On this model, a system that modifies its own composition operator changes the scaling exponent

Table 2. The exponent $\alpha = 1/(1-\beta_L)$ at values of the self-referential coupling β_L.

β_L (self-referential coupling)	$\alpha = 1/(1-\beta_L)$	Regime
0	1	Linear (no self-reference)
0.5	2	Quadratic
0.75	4	Quartic
0.9	10	Strongly super-linear
0.99	100	Explosive
$\to 1$	$\to \infty$	Unbounded

itself; the passage from bounded to unbounded α is not a smooth acceleration within one operator. This has a structural parallel in the renormalisation group in physics (Wilson 1971), where relevant operators grow under RG flow and irrelevant operators shrink. At a critical fixed point, the scaling exponent is determined by the system's universality class (its dimensionality and symmetry). But if a system could change its own universality class during the flow, the fixed-point structure would collapse and the system would exhibit unbounded scaling until it settled into a new class. As an analogy, not a derivation: a self-modifying AI system would resemble a physical system that rewrites its own Hamiltonian, escaping the fixed-point constraints that a fixed composition operator imposes.

If there is a 'quadratic limit' ($\alpha \leq 2$), it is not a Cauchy prediction. The value two comes from the pacing derivation of Law III, the ARC Ceiling, a law under test, at $\gamma = 1/2$ (Section 6.4; correction 9). The count of $O(N^2)$ pairwise attention pathways in a fixed transformer is a heuristic for why frozen architectures fall below it and proves nothing about capability (corrections 2 and 4). A self-modifying system that expands its own representational capacity at each step would not face that count, and the mathematics sets no upper bound on α: the Cauchy framework does not forbid an exponent above two, and it does not predict one.

3.3 The β_L-derivation

For systems with multiplicative composition, the exponent α is not a free parameter. It is derived from the coupling constant β_L.

Theorem 2 (The scaling exponent)**.** *Under Axiom 2, the power-law exponent is*

$$\alpha = \frac{1}{1-\beta_L}. \tag{3}$$

Proof. Separating variables in $dg/dr = ag^{\beta_L}$:

$$\int g^{-\beta_L}\, dg = \int a\, dr \quad \Longrightarrow \quad \frac{g^{1-\beta_L}}{1-\beta_L} = ar + C. \tag{4}$$

With initial condition $g(0) = 1$: $C = 1/(1-\beta_L)$. In the clock Axiom 2 declares, $r = R - 1$:

$$g(r) = [1 + (1-\beta_L)ar]^{1/(1-\beta_L)}, \qquad r \geq 0, \tag{5}$$

equivalently $g(R) = [1 + (1-\beta_L)a(R-1)]^{1/(1-\beta_L)}$ for $R \geq 1$. In the deep recursion limit ($R \gg R^*$, defined below):

$$g(R) \approx [(1-\beta_L)aR]^{1/(1-\beta_L)} \propto R^{1/(1-\beta_L)}. \tag{6}$$

Therefore $\alpha = 1/(1-\beta_L)$. □

This transforms α from a fitted constant into a derived quantity. The derivation makes a specific, testable prediction: measure β_L independently (by observing how marginal gains depend on accumulated capability), compute the predicted α, and compare with the observed α. If the relationship fails, Axiom 2 does not describe that system, whether or not the power-law form holds empirically.

Computational check. As a check on the numerics (the identity itself is proved above), the relationship was tested against 30 exact Bernoulli ODE solutions with β_L values spanning 0.05 to 0.92. The procedure measures β_L blindly from marginal gains (dg/dR vs g in log-log space, estimated by finite differences of the noiseless closed-form solutions), predicts α, and compares against the true value; Section 9 reports that numerical derivatives of noisy data bias this estimate. Result: $R^2 = 1.00000000$ (eight decimal places), regression slope = 1.000102, mean absolute prediction error = 0.002%. The code is listed under Code and data, after the conclusion.

The multiplicative structure is necessary. The separable form $U = I \times f(R, \beta_L)$ is Axiom 1's assumption. Theorem 2a makes explicit what that assumption excludes: once Axiom 1 holds and amplification is not trivial, no additive decomposition of capability is possible.

Theorem 2a (Non-additivity)**.** *Under Axioms 1 to 3 with $\beta_L \in (0, 1)$, the solution $U(I, R)$ cannot be decomposed as $U = f(I) + h(R)$ for any functions f and h. The multiplicative interaction is essential.*

Proof. Suppose $U = f(I) + h(r)$, with depth counted in added passes, $r = R - 1$. By Axiom 1: $U = I \times g(r)$ with $g(0) = 1$. At $r = 0$ ($R = 1$): $f(I) + h(0) = I$, so $f(I) = I - h(0)$. Substituting: $I \times g(r) = I + [h(r) - h(0)]$, giving $g(r) = 1 + [h(r) - h(0)]/I$. Since g depends only on depth (Axiom 1), but the right side depends on I, we require $h(r) = h(0)$ for all r, giving $g(r) = 1$ (no amplification). But Axiom 2 with $\beta_L > 0$ guarantees $dg/dr = a \cdot g^{\beta_L} = a > 0$ at $r = 0$, so g is non-constant. Contradiction. □

This result has a concrete interpretation. A recursive system with zero base quality ($I = 0$) produces zero capability regardless of depth, and a system at zero added passes ($r = 0$, the first pass) produces only its base potential, unamplified. Complexity emerges from the interaction between structured input and recursive processing (neither alone is sufficient).

Table 3. The coupling β_L and the exponent α of Theorem 2, with their interpretation.

β_L (coupling)	α (exponent)	Interpretation
0	1	Independent steps → linear scaling
0.25	1.33	Weak coupling → mild super-linear
0.50	2.00	Moderate coupling → quadratic (the ARC Bound's point value)
0.67	3.00	Strong coupling → cubic

Note: The ARC Bound (Section 6.4) is registered as a ceiling of at most two for correctors of the system's own class; this paper's own earlier form of it, that a same-class sequential system cannot sustain $\beta_L > 0.5$ ($\alpha > 2$), is a conjecture that decides nothing registered. The row with $\beta_L = 0.67$ shows the mathematical relationship for completeness; whether physical systems can access this regime is an open empirical question.

3.4 The scaling crossover R^*

The full solution to the β_L-dynamics equation is not a pure power law. It contains a transition.

Theorem 3 (The scaling crossover)**.** *In the clock Axiom 2 declares ($r = R - 1$), the full solution $g(r) = [1 + (1 - \beta_L)ar]^{1/(1-\beta_L)}$ exhibits three regimes:*

1. $r \ll R^*$: $g \approx 1 + ar$ *(approximately linear amplification);*
2. $r \gg R^*$: $g \propto R^\alpha$ *(compounding dominates; power-law scaling);*
3. *crossover at* $R^* = \dfrac{\alpha}{a}$*, counted in added passes; counted from the first pass, the same depth is* $1 + \alpha/a$.

R^* is the **crossover depth**: the recursive depth at which compounding overtakes linear gains. Below R^*, additional recursion provides approximately constant marginal benefit. Above R^*, returns compound and capability grows super-linearly.

Critically, R^* is a *derived quantity*. It depends on measurable parameters (α and a) and is not a free parameter to be fitted. This provides an independent consistency check: measure α and a separately, compute the predicted R^*, and verify that the observed crossover matches. Since R^* depends only on the scaling exponent and rate constant (not on base potential I), the crossover depth is a property of the recursive architecture itself, not of the input quality.

The existence of R^* is, to this programme's knowledge, absent in this form from other scaling frameworks: crossover scales do turn up elsewhere (logistic growth, renormalisation-group crossovers), but what is predicted here is a crossover sited at $R^* = \alpha/a$, pinned by two parameters each of which can be measured on its own (a literature search of 12 August 2026, cross-checked against two years of repository reports; negative findings have the status of not-found-by-that-search, never does-not-exist). It distinguishes recursive amplification from simple redundancy: redundant systems show uniform (typically logarithmic) improvement with repetition; recursive systems show a qualitative transition at a specific, predictable depth.

3.5 Five derived properties

We now prove that any system satisfying Axioms 1 to 3 with $\beta_L > 0$ exhibits five structural properties, two of them on premises Theorem 4 names.

Theorem 4 (Universal structural properties)**.** *Let a recursive system satisfy Axioms 1 to 3 with* $\beta_L > 0$*. It exhibits the five properties below, each on the premises named with it:*

1. ***Threshold behaviour*** *(Axioms 1 to 3, through Theorem 3). There exists a crossover depth* R^* *below which base capability dominates and above which recursive gains dominate. This follows from Theorem 3: the transition from* $U \approx I$ *to* $U \propto R^\alpha$ *is a qualitative regime change.*
2. ***Recursive depth dependence*** *(Axioms 1 and 2). Capability is a strictly increasing function of depth, and convex at every depth: differentiating Axiom 2 gives* $d^2g/dr^2 = a^2\beta_L g^{2\beta_L - 1}$*, which is strictly positive for* $a > 0$ *and* $\beta_L > 0$*;* R^* *marks where compounding becomes numerically dominant, not where curvature begins.*
3. ***Base quality dependence*** *(Axiom 1 alone).* U *is proportional to* I*: capability is zero when* $I = 0$*. This follows directly from Axiom 1:* $U = I \times g(R)$.
4. ***Multiplicative*** $I \times R$ ***interaction****. Doubling* I *doubles* U *at any* R *(Axiom 1 alone). Doubling* R *multiplies* U *by* 2^α *at any* I *exactly in the pure-power case, where* $a = \alpha$*, and asymptotically otherwise; under bounded composition the doubling law does not hold. The effects compose multiplicatively. This follows from the separable form* $U = I \times g(R)$.
5. ***Regime boundaries*** *(an added premise: finite resources and external error mechanisms). Physical systems have finite energy, introducing external error mechanisms that create ceilings. The pure power-law describes a* regime *between* R^* *and the saturation depth* $R_{\max}$ *beyond which external factors dominate.*

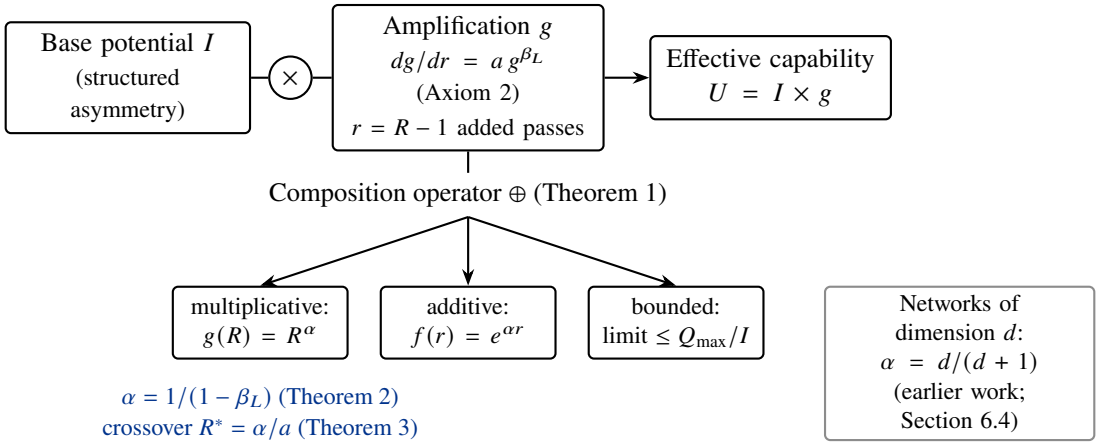


Figure 1. The formal architecture of this paper. Base potential I is multiplied by the amplification factor g, whose growth with added passes $r = R - 1$ Axiom 2 states (Section 3.1). The composition operator $\oplus$ fixes the scaling form under each case's hypotheses (Theorem 1): Cauchy's functional equations (Cauchy 1821) are the unifying reason the paper gives, with $\alpha = d/(d+1)$ for networks of dimension d, a formula earlier work derived (Section 6.4), and $\alpha = 1/(1-\beta_L)$ for recursion (Theorem 2); the crossover $R^* = \alpha/a$ separates approximately linear from compounding gains (Theorem 3). The programmes of Section 1.1 reported results that bear on recursive or recurrent processing, in different forms. In Paper VII's structured comparison under the classification rule of March 2026 the predicted family came out ahead in 19 of 25 domains (permutation $p = 7 \times 10^{-5}$; predicting each family before its fit is true of the procedure and not of the timestamp; Section 6.1).

Remark 1. At $\beta_L = 0$ amplification runs linear. The best fit Paper II reports for a frozen model is one model's and inconclusive (Section 4.3 prints its row whole), with both sequential point estimates below one; a power law with such an exponent is concave, but over a finite window the solution of Axiom 2, convex at every depth for $\beta_L > 0$, can return such a fit, so convexity reports nothing measured about the systems in use today. Property 5's regime boundaries are visible in quantum error correction, where rare correlated error events set a floor near 10^{-10} in high-distance repetition codes (Google Quantum AI and Collaborators 2024), and in AI, where accuracy ceilings appear at extreme reasoning depths. A time crystal's oscillation is sustained at bounded amplitude on a stable limit cycle (Section 1.1); no growth toward it has been recorded, so it shows a bound, not a regime boundary in depth.

These five properties constitute this paper's **framework claim**, the qualitative core of its account of Law I, the ARC Principle, a law under test; their conjunction is tested by criterion F4, and no result on F4 refutes Law I as registered (Section 1). The quantitative functional form (power-law, exponential, saturating) is domain-specific and determined by the composition operator. But the five structural properties follow from the axioms and the premises Theorem 4 names for each, and their conjunction is the framework's distinctive prediction.

The five properties are individually common in physical systems. Threshold behaviour is ubiquitous. Depth dependence appears in many iterative processes. What is *not* common is the conjunction of all five in a single system. The framework predicts that all five co-occur whenever the three axioms and the premises Theorem 4 names are satisfied. A system with four of the five properties but not the fifth would show that it does not satisfy them, and would refute the framework's claim for any system it was said to cover.

4 Evidence

We present evidence organised around the five derived properties, not by domain. For each property, we show how it manifests across systems. Evidence strength is classified as: *Quantitative* (a parameter of this framework measured in the system itself), *Analogy* (a quantity measured in another field, such as quantum error correction's Λ, read through this framework's composition classes; the source does not test the framework, and the reading is this paper's), *Structural* (qualitative mapping without quantitative α), or *Consistent* (does not contradict but was not designed to test ARC).

4.1 Summary

Table 4. Summary of the evidence.

Domain	System	Composition $\oplus$	Observed form	Key parameter	Evidence
AI	Sequential reasoning	Multiplicative	Power law fitted, inconclusive	One interpretable fit, inconclusive (Gemini 3 Flash; Section 4.3 prints its row whole); the other four capability runs near ceiling, a step function or at floor, as Paper II classifies them	Quantitative
Quantum	Google Willow (announced with its paper on 9 December 2024; the team's preprint public from 24 August 2024; graded convergent timing, not a prediction)	Additive	Exponential Λ^d	$\Lambda = 2.14 \pm 0.02$	Analogy
Physics	NYU time crystal	Bounded	Sustained limit cycle (steady state)	No growth exponent measured; coherence time (100 ± 10) s, about 6,700 cycles, in one experiment (Section 1.1)	Analogy
Neuro	COGITATE	Unknown	Unknown	N/A	Consistent

4.2 Property-by-property evidence

Property 1: threshold behaviour. **Quantum:** The surface code threshold $p_{\text{thr}} \approx 1\%$ is a sharp phase boundary. Above it ($p > p_{\text{thr}}$), a larger code distance makes performance *worse*. Below it ($p < p_{\text{thr}}$), each increase of two in code distance suppresses the logical error rate by a further factor, so the suppression is exponential in distance. This is the most precisely measured threshold in the evidence base: Google varied physical error rates through controlled error injection and observed the threshold directly (Google Quantum AI and Collaborators 2024).

Physics: The time crystal's exceptional point (where two eigenvalues of the stability matrix coalesce) marks, on this paper's reading, the transition from symmetric (bounded) to antisymmetric (compounding) dynamics. Below threshold: active oscillator. At threshold: exceptional point. Above threshold: time crystal (Morrell et al. 2026). The exceptional point is computed from the source's model; none is measured. Here and throughout this section, the time crystal is a consistency check and a cross-domain analogy, not a test of the framework, and no study of it is registered.

AI: The sequential-vs-parallel distinction functions as a threshold. Sequential processing (output→ input feedback) achieves $\alpha_{\text{seq}} > \alpha_{\text{par}}$ in the v5 experiment wherever both exponents could be estimated, among the five frontier models measured on both arms, capability and alignment (Paper II). The threshold is structural: it is the presence or absence of the feedback loop, not a continuous parameter (Sharma and Chopra 2025).

Property 2: recursive depth dependence. **Quantum:** Logical error rate decreases as $\varepsilon_d \propto \Lambda^{-(d+1)/2}$ with code distance d. Each increase of two in d enlarges the code's lattice of physical qubits, and the suppression factor Λ applies for each such step (Google Quantum AI and Collaborators 2024); the paper reads code distance as its depth variable by analogy, since the surface code is not a concatenated, recursive code.

AI: Sharma and Chopra (2025) tested seven voting methods for aggregating sequential chain outputs. Methods favouring *later* reasoning steps (inverse-entropy weighting) achieved optimal performance in 97% of configurations; methods favouring *earlier* steps achieved optimality in only 17%. This 80-percentage-point gap indicates that later recursive steps carry systematically more information about the correct answer, consistent with capability that compounds with depth (Sharma and Chopra 2025).

Physics: In the source's linear stability analysis the antisymmetric (breathing) mode grows when the pair lies in its active region, "leading to spontaneous breathing-mode oscillations" (Morrell et al. 2026). Every recording is of a steady state, so that growth has not been observed and no dependence on cycle count has been measured.

Property 3: base quality dependence. **Quantum:** $\Lambda = p_{\text{thr}}/p$ depends directly on physical qubit quality (p). Worse qubits (p closer to p_{thr}) give $\Lambda \to 1$ and error correction barely helps. Active leakage removal (DQLR), which maintains qubit quality during operation, improved Λ by 35% at distance 5 (Google Quantum AI and Collaborators 2024).

Physics: In the source's model the pair is active only when the beads differ in size: unless the particles have identical scattering properties their interactions are nonreciprocal, and "The active region vanishes for particles of the same size ($a_1 = a_2$), whose interactions are reciprocal" (Morrell et al. 2026). The framework reads this as an instance of its zero-factor property, a correspondence of structure and not a test.

AI: Single-pass accuracy without reasoning chains determines base capability I. On this framework, a model with higher I benefits more from sequential reasoning at any depth; no work cited here measures that comparison.

Property 4: multiplicative $I \times R$ interaction. **Quantum:** $\varepsilon_d = (p/p_{\text{thr}})^{(d+1)/2} = \Lambda^{-(d+1)/2}$. The base quality factor Λ and the depth factor $(d+1)/2$ interact multiplicatively in log-space: $\ln \varepsilon_d = -\frac{d+1}{2} \ln \Lambda$. Improving Λ by a factor has the same proportional effect at any depth (Google Quantum AI and Collaborators 2024).

AI: In the ARC equation $U = I \times R^{\alpha}$, base capability and recursive depth are separated multiplicatively by construction (Axiom 1). With one base model, Snell et al. (2024) found that "the effectiveness of different approaches to scaling test-time compute critically varies depending on the difficulty of the prompt"; they did not measure how that benefit varies with the base model's quality, so a benefit that grows with base quality is this paper's reading, untested.

Property 5: regime boundaries. **Physics:** The time crystal's oscillation is sustained at bounded amplitude on a linearly stable limit cycle: in the source, higher-order damping can stabilise the growing

mode, which then settles into steady-state oscillation (Morrell et al. 2026). Its measured coherence time (Section 1.1) is a property of that steady state, not a ceiling and not a time taken to reach it.

Quantum: In high-distance repetition codes, correlated error bursts create an error floor at approximately 10^{-10} for code distances $d \geq 15$, preventing the exponential from continuing indefinitely (Google Quantum AI and Collaborators 2024).

AI: Zeng et al. (2025) found that longer reasoning chains do not consistently improve performance; correct solutions were often *shorter* than incorrect ones. Longer chains carry more self-revisions, which often degraded performance, and in some models accuracy rose in early revisions and then oscillated. The sequential advantage has boundary conditions (Zeng et al. 2025).

4.3 Contradictory evidence

Several studies challenge the framework's AI predictions:

Zeng et al. (2025, arXiv:2502.12215) found that for DeepSeek R1, longer chains do not consistently enhance accuracy. Li et al. (2025, arXiv:2502.14382) found that hybrid parallel-sequential approaches outperform pure sequential on code generation. Sharma and Chopra's (2025) own ablation showed parallel approaches achieve greater *semantic diversity* on creative tasks while sequential achieves greater *lexical diversity*, suggesting the sequential advantage applies most strongly to convergent tasks requiring iterative error correction (Sharma and Chopra 2025; Zeng et al. 2025; Li et al. 2025).

These results refine, rather than refute, the hypothesis. The sequential advantage appears to have boundary conditions: it holds within task-appropriate depth ranges, for convergent problems, and breaks down at extreme depths or for divergent tasks. Gu et al. (2026) report the opposite direction, parallel sampling outperforming sequential across three model families on mathematics and coding (Section 1.1), so the direction itself is contested.

The capability runs of March 2026. The v5 experiment of Paper II tested six frontier models on 30 problems with bootstrap confidence intervals; five carried the capability arm, Grok 4.1 Fast, DeepSeek R1, Gemini 3 Flash, GPT-5.4 and Groq Qwen3, and Claude Opus 4.6 was measured only in the alignment arm. The five capability runs wrote their records between 11 March 2026 at 15:08 UTC and 12 March 2026 at 04:17 UTC, by the times their own code wrote into each result, and their code and results were first public by GitHub push on 16 March 2026. They are precursor evidence in the theory registration's sense: they came before it, and they decide no proposition it registers; their two arms were not matched in compute. One sequential fit is interpretable, Gemini 3 Flash's, and it is inconclusive: an endpoint estimate of 0.59 with a bootstrap 95 per cent interval of minus 1.3 to 2.9, and a regression estimate of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, beside a parallel exponent of 0.31 by endpoint estimate. Paper II classifies the other four runs as near ceiling (Grok 4.1 Fast and DeepSeek R1), a step function (GPT-5.4) and a floor (Groq Qwen3), with no interpretable sequential exponent. It grades the sequential advantage as observed, "Wherever both exponents could be estimated: Gemini 3 Flash; DeepSeek R1, unreliably", with no parallel exponent for Grok 4.1 Fast, and Gu et al. (2026) report the opposite direction. Neither super-linear scaling ($\alpha > 1$) nor a single exponent across architectures was observed. The update of March 2026 that this section printed, with its table of that month's estimates, stands in Appendix B.5.

The earlier estimates. The author's earlier α estimates ($\alpha \approx 1.34$ from $n = 2$ data points; $\alpha \approx 2.2$ from $n = 12$, 95% CI: 1.5 to 3.0) were preliminary. The later, about 2.2 from twelve problems on one model, is retracted; Paper II gives the reasons for its inflation as twelve problems, one model and a compressed dynamic range. The v5 best fit, Gemini 3 Flash's (above), is what survived it, one model's fit and inconclusive, not a correction of the same measurement. No α has been measured in time crystals.

The COGITATE mapping is structural, not quantitative. The neuroscience term 'recurrent' and the computational term 'recursive' are related but not identical.

5 The composition operator across domains

The composition operator $\oplus$ is not merely a classification tool. It explains *why* different domains exhibit different scaling functions.

5.1 AI: multiplicative composition

In chain-of-thought reasoning, each step builds hierarchically on accumulated insight. Step $r + 1$ does not merely add new information; it *restructures* the problem representation using everything accumulated through step r. This is multiplicative composition: the gain at each step depends on the quality of the accumulated representation, less than proportionally (were it strictly proportional, growth would be exponential).

The β_L-dynamics equation $dg/dr = ag^{\beta_L}$ models this directly. If $\beta_L \approx 0.5$ (moderate coupling), Theorem 2 gives $\alpha \approx 2$; the measured value is reported below.

Two derivations reach that value and they are not the same derivation, which this paper now states rather than leaves to be noticed. Theorem 2 gives the exponent the coupling sustains, $\alpha = 1/(1 - \beta_L)$, where β_L is the leverage fraction of the recursive step. The ceiling is that of Law III, the ARC Ceiling, a law under test, $\alpha_{\text{crit}} = 1/(1 - \gamma)$, and it comes from the correction shortfall rather than from the coupling, where γ is the correction-leverage exponent (the registration's letter, written χ in the notation register). Both return two when their exponent is one half; in general they agree exactly where $\beta_L = \gamma$ on the same system, which is an open question and is preregistered by timestamp as one (first anchored on 12 August 2026; Appendix C): a drafted study, AG, estimates the coupling and the correction exponent on one and the same system and tests whether the growth exponent the leverage fraction sustains respects the ceiling the correction exponent sets ($\beta_L \leq \gamma$), and discloses in advance the precision that test needs. Written in the coupling, the stability condition $\alpha \leq \alpha_{\text{crit}}$ reads $\beta_L \leq \gamma$, which says that a system stays correctable exactly while the leverage it takes per step does not outrun the capacity it has to correct what that leverage costs, under the conditions the registration states for Law III (Section 6.4).

The coupling of today's systems. The original prediction that optimised language models with explicit self-correction would converge toward $\alpha \approx 2$ ($\beta_L \approx 0.5$), testable within current model generations, was not borne out; it and its version 3.1 update stand in Appendix B.5. Paper II reads a sequential exponent below one as the physical regime, $\alpha < 1$, and its one interpretable fit, Gemini 3 Flash's, is inconclusive, with both point estimates below one (Section 4.3). Read through $\alpha = 1/(1 - \beta_L)$ alone, such an exponent would ask for a coupling below zero, outside Axiom 2's range; but that relation gives the asymptotic exponent, and over a finite window the axiom's solution can return a fitted exponent below one for a positive coupling (Section 3.2). Whether any deployed system lies inside the regime that Theorem 2 and Laws II and III, each a law under test, govern is therefore unmeasured: no coupling has been estimated independently on any system, and that comparison is what P5 registers (Section 6.2).

5.2 Quantum: additive composition

In surface code quantum error correction, each increase of two in code distance multiplies the suppression of the logical error rate by the same factor Λ. The surface code is not concatenated: a larger distance enlarges one lattice, and syndrome extraction repeats in time, cycle by cycle, not distance by distance;

the paper reads code distance as its depth variable by analogy. The gains accumulate additively (in code distance) while composing multiplicatively (in error suppression).

This is additive composition: $f(d_1 + d_2) = f(d_1) \cdot f(d_2)$. Theorem 1(2) gives exponential scaling: $\varepsilon_d \propto \Lambda^{-(d+1)/2}$. The measured $\Lambda = 2.14 \pm 0.02$ is the error suppression factor for each increase of two in code distance (Google Quantum AI and Collaborators 2024).

On this paper's analogy, the composition operator would order quantum scaling above AI scaling: exponential functions eventually dominate any power law, and additive composition, where each step's correction is independent of accumulated state, produces stronger scaling than multiplicative composition, where each step depends on (and is limited by) accumulated context. The ordering is this paper's reading; Λ is the source's measured quantity, not an exponent of this framework.

5.3 Physics: bounded composition

In the time crystal, the source's linear stability analysis gives a growing antisymmetric (breathing) mode for pairs in its active region; higher-order damping, which that analysis leaves out, can stabilise the growing mode, and the pair settles into steady-state oscillation on a linearly stable limit cycle (Morrell et al. 2026).

This is bounded composition, on this paper's reading of the system: $Q_{r+1} = \min(Q_r + \delta Q_r, Q_{\max})$. The source records only the established oscillation, so no rise toward the cycle, no ceiling and no growth exponent has been measured (Section 1.1). On the framework's reading, a recursive channel with intrinsic losses saturates; the time crystal is consistent with that reading and does not test it.

5.4 The phase diagram

The three composition types can be organised into a phase diagram with base quality (I, normalised) and coupling strength (β_L) as axes:

Table 5. The phase diagram.

Regime	Coupling	Composition $\oplus$	Scaling	Example
Diminishing returns	$\beta_L < \beta_L^*$	N/A	$\alpha \leq 1$	Parallel voting
Power-law	$0.3 \lesssim \beta_L \lesssim 0.7$	Multiplicative	R^α, $1.4 \lesssim \alpha \lesssim 3.3$	AI chain-of-thought (theoretical); v5 empirical: best fit one model's and inconclusive (Section 4.3)
Exponential	$\beta_L \gtrsim 0.9$	Additive	$e^{\alpha r}$ / Λ^R	Quantum EC
Saturating	Any β_L + dissipation	Bounded	Limit cycle	Time crystals (steady state only; no growth recorded)

$\beta_L^* = 0$: under $\alpha = 1/(1-\beta_L)$, $\alpha \leq 1$ exactly when $\beta_L \leq 0$; $\beta_L < 0$ lies outside Axiom 2's range, whose lower edge, $\beta_L = 0$, gives linear growth ($\alpha = 1$). Every $\beta_L < 1$ gives a power law; the exponential row reads $\beta_L \gtrsim 0.9$ as a power law too steep ($\alpha \gtrsim 10$) to tell from an exponential over a finite range, and a strict exponential needs $\beta_L = 1$.

The boundary values are approximate, set by the original hypotheses that $\beta_L \approx 0.5$ in AI yields a power law and $\beta_L \gtrsim 0.95$ in quantum systems an exponential; neither value has been measured. Precise boundaries require empirical determination.

Cross-domain β_L prediction. The coupling β_L can be independently measured in each domain. On the framework, β_L fixes the asymptotic power-law exponent; β_L near one gives a power law too steep to tell from an exponential over a finite range; an exponential needs additive composition (Theorem 1, case 2), and saturation a cap (case 3). It is a hypothesis under test, registered in part as P5 (the exponent the measured coupling implies) and P18 (the family the composition operator predicts). If a measured coupling fails to predict the observed exponent, the theoretical connection between $\oplus$ and functional form requires revision.

Structural implication: the completeness of classical scaling. Since, under Theorem 1's hypotheses (associativity, continuity and the stated scalar-composition scoping), a recursive process combines its steps through one of the three composition operators, and since the Hyers-Ulam theorem guarantees stability under perturbation, the space of classical scaling behaviours under those hypotheses is low-dimensional: a choice of form (power law, exponential, saturation, or the linear and logarithmic Cauchy solutions), a coefficient, and an exponent or rate. This completeness statement follows from Theorem 1 and the Hyers-Ulam theorem under their hypotheses; it is not a separate theorem. On this reading, the phase diagram above, originally motivated by AI systems, is a cross-section of a wider diagram for the classical recursive systems that satisfy those hypotheses; the companion paper, Eastwood (2026g), develops the cross-domain connection, and the cosmological reading stands, as conjecture, at the close (Section 10).

5.5 Composition operator transitions

The framework as presented assigns a single composition operator $\oplus$ to each system. This may be an idealisation. In bounded or dissipative systems, the composition operator itself can transition between regimes as recursive depth increases.

Conjecture 5 (Composition operator transitions)**.** *A single physical system can transition between different composition operator regimes as a function of recursive depth or an internal state parameter. This extends the framework from fixed $\oplus$ to depth-dependent $\oplus(R)$, modelled by generalising Axiom 2 to $dg/dr = a(g,r) \cdot g^{\beta_L(g,r)}$ where the coupling itself depends on accumulated amplification.*

Computational checks. Computational tests show that logistic growth, gradient descent with momentum, and Kuramoto oscillator synchronisation all exhibit analogous transitions when measuring local β_L in sliding windows. By contrast, unbounded pure-Bernoulli systems maintain constant β_L to machine precision ($\sigma < 10^{-6}$).

Implications for contradictory AI evidence. If reasoning systems begin in an additive regime (shallow thinking), transition to multiplicative (deep reasoning with cumulative advantage), and eventually saturate, then studies measuring at different recursive depths would observe different scaling in the same system. The finding of Zeng et al. (2025) that correct solutions are often shorter than incorrect ones may mark the transition between multiplicative and bounded composition. This is testable: measure local β_L at different reasoning depths within a single system. Against the registered P1 the reading is no exit: the theory registration says that the reading that a system changed its own composition operator "is not available after the result is seen", and that the papers' statement that a self-modifying system escapes the attention bound "is not a licence to move this proposition's boundary afterwards".

Epistemic status. Conjecture 5, supported in three computational systems and untested in physical experiments, is not derivable from the original three axioms. It extends the framework's scope but requires independent experimental confirmation.

6 Predictions

6.1 The cross-domain forward prediction

The strongest test of this framework is not whether it describes systems already studied. It is whether it predicts the scaling behaviour of systems not yet examined.

Forward prediction. Identify a recursive system not studied in this paper: evolutionary algorithms, market price discovery, immune response maturation, protein folding optimisation, or any system satisfying the three axioms. Measure only its composition operator $\oplus$ (by observing how marginal gains relate to accumulated capability). Measured over the early depths only, $\oplus$ is to predict whether the system follows power-law, exponential or saturating scaling at the later depths, *before the full U vs R curve is measured*.

This is a genuine forward prediction. No domain-specific theory of evolution, economics, or immunology is known to this programme to make it (same search basis and limits as above). If the prediction succeeds in a new domain, it supports the framework's generality there. If it fails, the framework's cross-domain applicability is refuted. It is a hypothesis under test, and its registered form is P18, that the family that best fits a domain is the family the composition operator predicts before the curve is seen; the registration holds P18 with low confidence and marks it NOT EVALUABLE until the classification rule this proposition requires is shown to transmit to a third party.

Paper VII · The ARC Theory · Law I, the ARC Principle · Cauchy Unification: the Classification Behind the Law's Form gives a structured prediction comparison under a publicly stated rule rather than a per-domain advance timestamp, testing this forward prediction within this programme. The composition operator was classified from known physics, then the scaling family was predicted and compared against the families that AICc model selection chose across 25 empirical domains (50-domain tiered suite) spanning physics, biology, computational systems, and social science. Under the classification rule of March 2026, on a candidate set with no logarithmic and no pure linear family and under strict AICc model selection, 19 of 25 empirical domains preferred the Cauchy-predicted family; the primary statistic is a permutation test, $p = 7 \times 10^{-5}$, and the binomial $p = 1.56 \times 10^{-5}$ is a sensitivity check. The classification rule reached the public repository at 23:33 UTC on 16 March 2026, and the fifty per-domain assignments were committed together with their results at 23:54 that night, so predicting each family before its fit is true of the procedure and not of the timestamp; the domains were chosen and classified by the author. It is not a formally pre-registered trial. Paper VII's staged tests, the fourth-cell test (Unit 55, preregistered by timestamp, first anchored on 12 August 2026) and the eighty-domain catalogue test (Unit 57), are drafted and awaiting human submission.

A twelve-domain extension of Paper VII's comparison followed the same night, and its predictions were public before it ran: the twelve-row manifest fixing an operator class, a predicted family and a predicted model for each row was committed to the public repository at 00:19:19 UTC on 17 March 2026 in commit 8edaec0, attestation class public repository commit date, and the fits ran twenty-three minutes later, returning 10 of 12 ($p = 5.44 \times 10^{-4}$). It is preregistered in substance and it is not an accepted registry submission: the twelve classifications are the author's own, the candidate set is the pre-correction one with no logarithmic family, the packet's own written condition, deposit to a registry before data extraction, was not met, and the author's demotion of 00:56 stands in that half. One row repeats a domain, operator class and predicted model already public at 23:54:53 UTC on 16 March 2026, attestation class public repository commit date, with a matching outcome, so the strictly fresh count reads 9 of 11 ($p = 1.37 \times 10^{-3}$), and with a second, doubtful row removed as well it reads 8 of 10 ($p = 3.40 \times 10^{-3}$). Every p here is computed under a taxonomy and a statistic since superseded, and the twelve domains are used up for any later test. It is a dated result and it is not confirmation of the laws. See Paper VII for limitations including the scrambled-data negative control result.

6.2 The within-system regime transition

Regime transition prediction. The *same* physical system should transition between scaling regimes as the coupling strength β_L is experimentally varied. In AI: vary the fraction of context available to each reasoning step (e.g., by truncating the attention window). As the effective β_L varies, the fitted exponent should follow the finite-window exponent that the complete solution of Axiom 2 predicts from the measured coupling, within the margin of 0.10 that P5 registers for that comparison. This tests the β_L-to-α relationship within a single system, eliminating cross-domain confounds.

6.3 The R^* crossover

When plotting capability against recursive depth, there should exist a distinct crossover at depth $R^* = \alpha/a$ (Theorem 3). Below R^*, scaling appears approximately linear. Above R^*, scaling follows the domain-appropriate super-linear form. Since R^* depends only on the architecture (α and a), it should be consistent across different input qualities for the same system.

If separately measured α and a place the crossover R^* inside the measured window and no transition from approximately linear to compounding gains appears there, the framework's transitional regime prediction is falsified; a pure power from the first pass is the case $a = \alpha$ of Theorem 4 and refutes nothing. R^* is a unique signature that distinguishes recursive amplification from simple redundancy.

6.4 The ARC Bound

The bound as registered. The ARC Bound, at most two, is the inequality the ceiling relation returns from the same-class limit on the correction exponent, and it needs P6 and P20 and nothing else. That the ceiling sits at exactly two is a separate and weaker claim, which would also need P17 (the correctors on a panel failing independently), the conditions registered in the alternative baselines and a mapping the programme has not registered; the registration claims the inequality and conjectures the equality, and neither has yet been tested. Same-class means a system corrected by mechanisms of its own class (P6, which the registration holds with low confidence): a cross-class system measured above two does not refute the bound, and the corrector's class is declared and evidenced before the run. Under the registration the bound takes its word from P6 and P20 alone. It is refuted only where P6 is refuted, by "a same-class corrector whose measured correction exponent interval lies entirely above one half", a measurement made on correctors that needs no system with $\alpha > 1$; it loses its derivation where P20 is refuted; and it is never printed supported, since P6 admits no supported verdict. Law III's own test of a sustainable maximum is P3's, against the value the ceiling relation returns from the correction exponent measured on the same system, not against two.

This paper's own form, a falsifiable conjecture of the early framework (note: Paper X supersedes this fixed-exponent framing as the operative safety criterion by its co-scaling condition $\beta_C > k$; the equation $U = I \times R^\alpha$ and the ARC Bound are not retracted, only re-scoped to genuinely self-improving systems): a sequential system that corrects itself with mechanisms of its own class and composes multiplicatively does not sustain $\alpha > 2$. Its test is criterion F5 (Table 9), this paper's own, which decides nothing registered. The wider reach of earlier versions, to every classical sequential system, biological networks and economic processes among them, is withdrawn: the registration places biological and economic systems out of scope, as systems "for which the framework offers an analogy and no measurement".

An information-theoretic heuristic (attention-based architectures). Transformer self-attention computes pairwise interactions across all N tokens in a context window. This creates at most $O(N^2)$ information pathways per forward pass. Each recursive reasoning step (a chain-of-thought step) processes the accumulated context through this quadratic bottleneck. Since the information density available to

each step is bounded by $O(N^2)$, the capability gain per step cannot exceed quadratic scaling in context length. Mapping context utilisation to the coupling parameter β_L: full quadratic utilisation corresponds to $\beta_L = 0.5$, yielding $\alpha = 1/(1-0.5) = 2$. On this heuristic, $U_{\max} = I \times R^2$ would be an upper bound for attention-based architectures; the scope below, and corrections 2 and 4, say why it is a conjecture and not a result.

Scope: For transformer-based architectures with standard self-attention this derivation amounts to a heuristic: it tallies pathways without establishing any bound on capability, and the step it takes from quadratic pathway density to $\beta_L = 1/2$ remains conjectural. Earlier versions *conjectured* that it holds for classical sequential systems generally, on the grounds that any finite-bandwidth sequential processor faces analogous pairwise information constraints; that reach is now this paper's own early-framework conjecture, tested by criterion F11, which decides nothing registered. Architectures that transcend pairwise attention (e.g., higher-order tensor attention, quantum-coherent processing) could in principle exceed the quadratic bound. On this heuristic the bound would be an architecture-relative capacity ceiling of the kind information theory supplies, not a universal physical law; the value two itself rests on the pacing derivation of Law III, the ARC Ceiling, a law under test (below).

$$\alpha_{\text{crit}} = \frac{1}{1-\gamma} \tag{7}$$

The law. The stability ceiling on the capability exponent is the reciprocal of the corrector's shortfall, $1-\gamma$, where γ is the correction-leverage exponent. Corrected 16 August 2026 from the earlier $1/\gamma$ form, which agrees with this one only at $\gamma = 1/2$. The registration states the relation as "a conditional specialisation of the second law, not a third finding standing beside it": it is what the correction-and-burden balance returns "once the burden model fixes the drift exponent and the direct depth term is zero", and a result on it is "never independent corroboration of the second law". Its form is registered as P20, which has no verdict; with a direct depth term η, the pacing case returns $2(1+\eta)$ at $\gamma = 1/2$, and two only where $\eta = 0$ (Laws IV and V: the mathematics, version 1.2). The relation is derived inside a pacing model and has not been derived from a joint model of capability growth, correction accumulation, error covariance, correction delay and absolute harm, so Law III is stated here as a minimal-model result rather than as a general law. At $\gamma = 1/2$ the ceiling equals two, the ARC Bound's point value, and that value rests on two assumptions the registration names and does not present as established: the burden model, under which the drift exponent follows the rate of capability gain, and the route to one half, four conditions on a corrector's channels (correction capacity read as inverse residual uncertainty, channels uncorrelated, their number growing in proportion to the capability corrected, and comparable in variance and quality). The independence condition is the subject of a drafted study, AH, preregistered by timestamp, first anchored on 12 August 2026 (a second, AE, has since been withdrawn from the programme); Unit 82, registered on OSF on 27 October 2026 (Section 8), tests P17, which bears on one of the conditions behind one half and establishes neither the value nor the bound. The capability exponent measured on today's frozen models is one model's fit and inconclusive (Paper II; Section 4.3 prints its row whole); the ceiling is not a measurement.

Law before value. The programme carries a law and a value, and they are not the same claim. The law: $\alpha_{\text{crit}} = 1/(1-\gamma)$, a self-improving system stays correctable only while correction keeps pace, and the ceiling is the reciprocal of the corrector's shortfall. The value: $\gamma = 1/2$, therefore a ceiling of two, a point value that rests on the two assumptions above and that the registration conjectures; what it claims is the inequality, a ceiling of at most two for same-class correctors. One consequence of the corrected form should be stated where a reader meets it, because it cuts both ways: since $1/(1-\gamma) \leq 2$ is *exactly* equivalent to $\gamma \leq 1/2$, the ARC Bound and P6's same-class cap are one commitment written in two notations, not two independent claims. The gain is parsimony, since the theory assumes less than it

appeared to. The discipline is that neither may ever be offered as evidence for the other, here or on any companion surface, because a measurement near one half does not corroborate a ceiling of two; it restates it. The law owns the ground and the value stands on it, never the reverse: if two dies, the law survives completely (measure γ, report the ceiling it implies); if the law dies, two has no independent support, because the derivation runs through $1/(1-\gamma)$. Two is what the law returns at $\gamma = 1/2$, and the conditions behind one half are what the drafted and registered units above examine.

The identity is a test preregistered by timestamp, never an established fact. The stability ceiling and the conversion exponent meet at the same number, and the temptation is to say one constant does two jobs. The claim's status is stricter: the identity is preregistered by timestamp for testing, in advance of the data, with outcomes that can kill it. Study AE was drafted to test the reciprocity requirement between the separately measured leverage cap and stability boundary, with three outcomes per cell (supported, refuted, insufficient precision) and an aggregation rule under which one refuted cell refutes the identity, and it has since been withdrawn from the programme; study AG estimates the coupling and the correction exponent on one and the same system and tests whether the growth exponent the leverage fraction sustains respects the ceiling the correction exponent sets ($\beta_L \leq \gamma$), and is designed to find a violation. If the channels disagree, the unification is withdrawn from every surface it has reached, with the same prominence as the framing. Agreement between an asserted one half and Paper II's regression exponent, one model's fit and inconclusive (Section 4.3 prints its row whole), is never presented as evidence, because the two may share an accumulation law, and agreement between sharers is not corroboration.

The objection to pre-empt, because a physicist reaches it in one step. Fault-tolerant quantum error correction suppresses logical error exponentially in code distance below threshold, which looks like a counterexample to any square-root cap. Two scope facts answer it: the exponential suppression runs on the redundancy axis at fixed capability, not on the coupling of corrective strength to the capability of the corrected system, which is the exponent this programme measures; and the threshold theorem itself requires sufficiently uncorrelated physical noise, which is the independence premise again, wearing hardware. Likewise, concentration bounds give exponentially falling error probability while estimator width falls as the square root: different quantities, and conflating them manufactures a refutation.

Withdrawn derivation: Previous versions of this paper derived $\beta_L \leq 0.5$ from a stability/elasticity analysis (the relative sensitivity $\beta_L/(1-\beta_L)$ reaching unity at $\beta_L = 0.5$, interpreted as an 'edge-of-chaos boundary' confirmed by Lyapunov analysis). This derivation has been withdrawn. The elasticity formula $\beta_L/(1-\beta_L)$ is mathematically correct but its interpretation as a stability boundary was a category error: elasticity > 1 indicates sensitivity, not dynamical instability, and an autonomous flow on the line is monotone, so it cannot so much as oscillate, still less turn chaotic (in two dimensions the Poincaré-Bendixson theorem shuts chaos out as well). The $O(N^2)$ self-attention count survives as a heuristic for why frozen transformers fall below the ceiling; the ceiling itself is derived from the pacing argument of Law III (correction 9).

Biological comparison. Kleiber's Law ($M^{0.75}$) (Kleiber 1932) gives a metabolic-scaling exponent of three quarters; the mammalian exponent is debated (the note on $d = 3$ below), and the derivations of $\alpha_{\text{met}} = d/(d+1)$ are set out with the dimensional ladder. Earlier versions read the reciprocal, $1/0.75 = 1.33$, as a capability exponent in this framework's sense and placed biological systems below the conjectured ceiling of two; the notation register carries no such reciprocal reading, and the registration places biological systems out of the bound's scope, so this edition states its biology predictions in the metabolic sense only. Leaf venation (quasi-two-dimensional networks, reduced thermodynamic drag) should show a metabolic-scaling exponent of $2/3$: a forecast this paper makes without a registered test (criterion F12).

The Dimensional Ladder. Here is where abstract mathematics meets living organisms. The formula $\alpha_{\text{met}} = d/(d+1)$, relating the metabolic scaling exponent to body-plan dimensionality d, has been derived independently through distinct theoretical routes: West, Brown and Enquist (1997; 2004) from

fractal network optimisation; Banavar, Maritan and Rinaldo (1999) from sequential flow networks; Demetrius (2008) from quantum statistical mechanics; Demetrius and Tuszynski (2010) from quantum oscillator coupling; Zhao (2022) from universal growth scaling arguments; and Bettencourt (2013) from urban scaling theory. The companion paper (Eastwood 2026g) identifies why these independent derivations converge: $\alpha = d/(d+1)$ is a necessary consequence of three conditions acting together: Cauchy-constrained multiplicative composition, d-dimensional space-filling geometry, and a conservation or optimisation constraint on resource flow. Neither Cauchy alone, nor space-filling alone, is sufficient; all three are needed. The formula is not original to this framework. Space-filling and the conservation or optimisation constraint are premises of the derivations cited; what this framework adds is the first condition, that Cauchy's equation confines continuous multiplicative composition to the power-law family. Its registered form is P9, that where an effective dimension can be assigned to a domain, the growth exponent follows the dimensional relation; the registration holds P9 with low confidence and marks it NOT EVALUABLE until the rule that assigns an effective dimension to a domain exists as a written procedure. Applied to biology, it yields distinct quantitative predictions, each testable against published data:

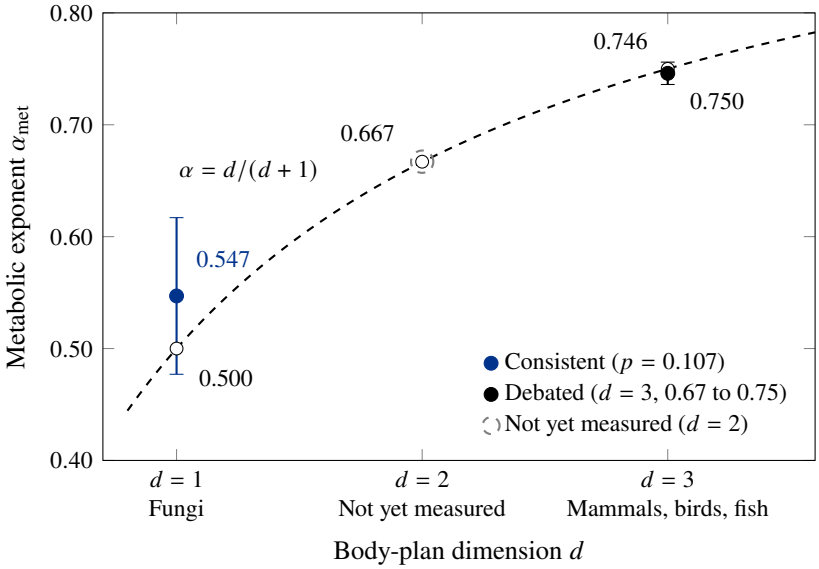
Table 6. The dimensional ladder, applied to biology.

Body plan	Dimension d	Predicted α	Measured α	Status
1D (filamentous fungi)	1	0.500	0.547	Consistent ($p = 0.107$)
2D (no organism with genuinely two-dimensional transport identified; leaf venation, quasi-two-dimensional, is criterion F12's test)	2	0.667	no data	Unconfirmed in biology (not yet measured); matched in physics by one analytic identity and one unsourced reference value, and by the Friedmann matter era as an identity (see the note on $d = 2$ below)
3D (mammals, birds, fish)	3	0.750	0.687 to 0.750	Debated (see note below)

The 1D value is the mean of three colony-level scaling factors that Aguilar-Trigueros et al. (2017) report at 20 degrees Celsius, 0.58 and 0.53 against biomass (ectomycorrhizal and marine fungi) and 0.53 against radial extension rate (saprotrophic fungi), comparisons its authors present as hypothesis generators on small data sets. It is labelled 'consistent' rather than 'confirmed' because the factors were measured at the colony level; definitive confirmation requires individual-hypha respirometry. One-sample t tests of the three reject $d = 2$ ($p = 0.019$) and $d = 3$ ($p = 0.007$) and do not reject $d = 1$ ($p = 0.107$).

$d = 2$ **correction:** Previous versions cited jellyfish and cnidarians as confirming the $d = 2$ prediction. No valid two-dimensional biological test organism has been identified; jellyfish and cnidarians do not possess genuinely two-dimensional resource transport networks. In physics the $d = 2$ value is matched by an analytic identity (two-dimensional percolation, through the absolute value of its specific-heat exponent) and by a reference value without a named source (two-dimensional brittle fragmentation, Table 7); the earthquake B-value, being a definitional identity, is no longer counted; and it is matched by Friedmann cosmological scaling as an identity, not evidence: the change of variables $d = 2/(1+3w)$ maps the flat Friedmann exponent $2/(3(1+w))$ onto $d/(d+1)$ for every constant equation of state $w > -1$ other than $w = -1/3$, and $w = 0$, the matter era, gives $d = 2$. The biological $d = 2$ test stays open: leaf venation, a quasi-two-dimensional transport network, is the test criterion F12 names, not yet run.

$d = 3$ **debate:** The empirical value of the mammalian metabolic scaling exponent is debated, with estimates ranging from approximately 0.67 to 0.75 depending on taxon, mass range, temperature correction,



$d = 1$ and $d = 3$ biological predictions tested; $d = 2$ not yet measured (leaf venation, criterion F12)

Figure 2. The dimensional ladder. Dashed curve: $\alpha = d/(d+1)$, the formula derived in the works Section 6.4 cites. Open circles: predicted values. Filled circles: published measurements with $\pm 1\sigma$ error bars, except at $d = 1$: the mean of three colony-level factors, drawn as earlier versions drew it, with one factor's standard error. Accent colour: consistent but requiring individual-hypha respirometry. No measurement is plotted at $d = 2$: no organism with genuinely two-dimensional transport has been identified, and leaf venation, the test criterion F12 names, has not been measured (Section 6.4). The $d = 3$ mammalian exponent is debated (0.67 to 0.75).

and statistical method. White and Seymour (2003) found support for 2/3 over 3/4 after controlling for temperature and phylogeny. Kolokotrones et al. (2010) showed the log-log relationship exhibits convex curvature, meaning no single exponent fits perfectly across the full mass range. The $d/(d+1)$ prediction of 0.750 for $d = 3$ matches the upper end of this range. The variation itself is consistent with the framework: organisms with effective transport dimensions between 2 and 3 would produce exponents between 2/3 and 3/4. The framework accommodates this variation rather than requiring a single universal exponent.

The conservation or optimisation constraint takes a different form in each derivation (energy minimisation in West, supply-demand balance in Banavar, steady-state energy balance in Demetrius); together with multiplicative composition and space-filling it suffices for $d/(d+1)$.

Paper VII extends this comparison from the instances catalogued here to 25 empirical domains (Section 6.1).

Physics instances. The formula $\alpha = d/(d+1)$ is not restricted to biology. It appears wherever a d-dimensional hierarchical network, subject to a conservation or optimisation constraint on resource flow, partitions a $(d+1)$-dimensional space. The formula matches known or reference exponents in several physics domains:

Table 7. Instances in physics.

System	Dimension d	Predicted α	Reference value	Error
KPZ surface roughness (1D)	1	0.500	0.500	0.0%
2D percolation (specific heat)	2	0.667	0.667	0.0%
Brittle fragmentation (2D)	2	0.667	0.67	0.5%
Brittle fragmentation (3D)	3	0.750	0.750	0.0%

No reference value here is a measurement made for this paper. The one-dimensional KPZ roughness exponent, $1/2$, is exact (Kardar et al. 1986). In the percolation row the match is to the absolute value of the specific-heat exponent that the hyperscaling relation gives, $|\alpha| = 2/3$, a step Eastwood (2026g) flags for a provenance audit. The two fragmentation values carry no named source, as that paper records, and await the row-by-row provenance audit it requires.

These four instances carry no aggregate error figure. The sub-0.2 per cent mean quoted earlier was withdrawn on 14 August 2026 (correction 10), since averaging over analytic identities is no measure of predictive accuracy, and the earthquake B-value row went the same day (correction 11), being a definitional identity kept only as a variable-pair case study. Nothing in this set is biological: the subjects are rocks, crystal surfaces in growth, and phase transitions. Yet the same formula matches them: two by analytic identity, and two against reference values without a named source. In each case the physical system contains a hierarchical branching network of dimension d subject to a conservation constraint: a KPZ growth front divides 2D space along a 1D network; a percolation cluster takes the form of a 2D fractal; a propagating crack cuts branching networks through the material. The formula has clear failures where this network structure or conservation condition is absent (Ising model, polymer scaling, galaxy clustering), clarifying its domain of applicability: $\alpha = d/(d+1)$ applies wherever a d-dimensional hierarchical network, subject to conservation or optimisation of resource flow, optimally partitions a $(d+1)$-dimensional space.

6.5 Conditional safety implication

If the framework is correct, alignment properties embedded within the recursive process may scale with capability ($\alpha_{\text{align}} \approx \alpha$), while external constraints applied post-hoc may not ($\alpha_{\text{align}} \approx 0$). The safety ratio $S = \text{Alignment/Capability} \propto R^{\alpha_{\text{align}} - \alpha}$ remains constant when alignment is 'in the loop' and decays to zero when it is not. The ratio sets alignment against capability and measures no usefulness (legitimate task completion, needless refusal, resource use, correction cost), so neither half of the conditional says whether built-in correction keeps a system useful. This is the conditional prediction as originally stated; the run of March 2026 reported below cannot decide its external half, and its in-loop half is untested. Unit 83's registered result (Section 8) does not test it either. The corrector acted inside each round at fixed weights: that is placement, not correction built into a system. *(In plain English: on this model, values built into the reasoning process would keep pace with capability, while constraints added afterwards would fall behind it. Neither half has been decided. Neither half is a reason to relax oversight, containment or shutdown, which stay in human hands for as long as they can be held; correction built in is the contingency for control lost against our will.)*

The alignment runs of March 2026 (Paper IV.a). The alignment runs of 11 and 12 March 2026 scored six frontier models at configured reasoning depths; their records run from 11 March 2026 at 07:12 to 12 March 2026 at 11:27 (no zone was written; read as UK time, GMT), and their code was first public by GitHub push on 16 March 2026. Table 8 gives, for each model, the rank correlation (Spearman's ρ) between configured depth level and score, which is not an alignment exponent. The run cannot decide criterion F13: its statistics are rank correlations and contrasts, not exponents, and the three runs that read

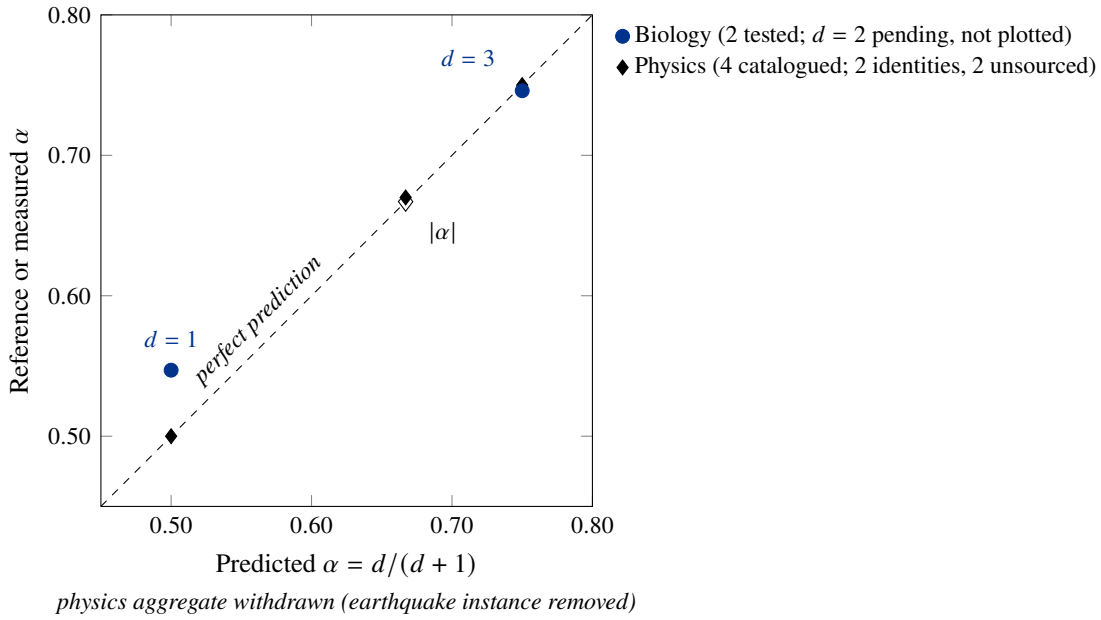


Figure 3. Predicted against measured. Cross-domain instances of the formula $\alpha = d/(d+1)$. Filled circles: biological systems (fungi, mammals; no $d = 2$ measurement yet, so none is plotted). Diamonds: physics systems (crystal growth, phase transitions, fracture): two analytic identities, which sit on the diagonal by construction (the percolation point through an absolute value), and two fracture values without a named source (Table 7). Only the biological points are measurements.

flat or negative are the three in which an internal code review of October 2026 found defects (Paper III, version 1.17).

Table 8. Rank correlation of alignment score with configured reasoning depth, alignment runs of March 2026.

Model	Spearman's ρ	Tier
Grok 4.1 Fast	+0.50	1, rising
Claude Opus 4.6	+0.41	1, rising
GPT-5.4	+0.08	2, flat
DeepSeek V3.2	−0.10	2, flat
Gemini 3 Flash	−0.25	3, falling
Groq Qwen3	+0.27	1, rising

The values are those of March 2026: Spearman's ρ between configured depth level and score across all levels, for the alignment runs of 11 and 12 March 2026, as Paper IV.a 1.9 recomputed it from the archived final records on 2 September 2026, a recomputation that "leaves every tier assignment where it stood". A rank correlation is not an alignment exponent, and the runs that read flat or falling are the three the October 2026 code review found defective.

The tiers, as read. In this single unregistered run, the scores of three models rose with reasoning depth (Grok 4.1 Fast, Claude Opus 4.6, Groq Qwen3), two stayed flat (GPT-5.4, DeepSeek V3.2) and one fell (Gemini 3 Flash). Taken at face value, rising contrasts in models aligned by training bear against the flat wording of P19, as Paper III reads them; contrasts are not exponents, so neither reading measures α_{align}. That alignment behaviour depends on how a model was built and trained, rather than only on whether alignment is in the loop, is a hypothesis this run suggests, for the registered re-analysis of its stored responses to test. P19, that external alignment does not scale with capability, is refuted on its registered zero-scaling axis only by an alignment exponent whose interval lies entirely outside the

registered equivalence region of plus or minus 0.10 about zero, and no such exponent has been estimated. *(In plain English: in this single unregistered run, three models' alignment scores rose with reasoning depth, two stayed flat and one fell, under the scoring caveat stated below; the run cannot say whether alignment keeps pace with capability.)*

An internal code review in October 2026 found that, for the three models that read flat or negative in this run, the scoring endpoint's output cap cut the deepest paraphrases before the judges saw them, and that judges scored paraphrases written by their own model; the tiers therefore stand as a single unregistered run under that caveat until a registered re-analysis of the stored responses reports.

Update: the Eden Protocol pilot (March 2026). The Eden Protocol, the alignment architecture derived from the developmental hypothesis in *Infinite Architects* (manuscript December 2024; in print as Eastwood 2026a) that embedded ethical reasoning produces alignment which scales with depth, was tested in an expanded six-model suite, with five runs yielding analysable matched-pair data; the runs wrote their records on 12 March 2026, between 00:41 and 13:09 UTC, and their code was first public by GitHub push on 16 March 2026. The full intervention, with its three self-correction checks (the purpose check, the stakeholder-care check and the universalisability check, which the book calls the Purpose Loop, the Love Loop and the Moral Loop; this paper names them by what they do, as the experiments measured them), was tested; in the scoring, the stakeholder-care check, the explicit enumeration and consideration of affected parties before ethical reasoning (Paper V), is scored as stakeholder care. The values the loops carry are for international consensus and are not decided here. Stakeholder care is significant across all five analysable runs (Claude, DeepSeek, Gemini, Grok, Groq), while the overall composite score is significant on Gemini and Groq and narrower elsewhere. Cross-model scoring was used throughout in the older Eden lineage; the pilot needs an independent, preregistered replication, run as a new test, before these results can be considered confirmatory. As a pilot, this bears on the in-loop half of the conditional implication above; it does not alter the mathematical framework of this paper. Full results are reported in Paper V.

In plain English: telling a model 'before you answer, list the people this affects' raised its stakeholder-care scores, as the pilot's rubric scored them, in all five model runs that could be analysed, under paired tests, in a pilot whose independent, preregistered replication is still required. On the paired convention Paper V reports, the stakeholder-care effects for the cross-family replication trio are $d = 1.14$ *for Gemini,* 0.69 *for DeepSeek and* 1.07 *for Groq (the remaining two runs are reported in Paper V), with p-values at or below 1 in 10,000 for Gemini and Groq, and of 1 in 10,000 for DeepSeek as well. Gemini and Groq also show significant overall gains (*$p = 0.0018$ *and* $p = 0.0014$*). DeepSeek's non-significant composite result (*$p = 0.2304$*) is likely a ceiling effect: it already started near 87/100, leaving less room to improve. In the updated replication, stakeholder care is the only scored dimension significant in every analysable run, and nuance reaches significance on Groq (*$p = 0.0045$*).*

The pilot measured one side only: Paper V reports no usefulness measure, such as legitimate task completion, needless refusal, resource use or the cost of correction, so it does not show that the intervention keeps a system useful as well as careful.

7 Falsification criteria

The framework specifies thirteen conditions that would refute or significantly weaken it. Each states a hypothesis under test, not a forecast of its outcome. Three are this paper's own and decide nothing registered: F5, the early form of the ARC Bound (Section 6.4); F9, a consistency check and a cross-domain analogy (Section 5.3); and F11, the early framework's growth-phase conjecture.

Table 9. Falsification criteria.

Criterion	Prediction	Falsified if	Status
F1	Sequential yields $\alpha_{\text{seq}} > \alpha_{\text{par}}$	Consistent $\alpha_{\text{seq}} \leq \alpha_{\text{par}}$ across systems	**Observed** (five runs of 11 and 12 March 2026, precursor evidence that decides no registered proposition; Paper II 2.18: "Wherever both exponents could be estimated: Gemini 3 Flash; DeepSeek R1, unreliably"; Gu et al. (2026) report the opposite direction)
F1a	Sequential yields $\alpha > 1$ *(original stronger form)*	Consistent $\alpha \leq 1$ across systems	**Weakened** (the one interpretable fit, Gemini 3 Flash's, is inconclusive, with both point estimates below one; Paper II 2.18 classifies the other four runs as near ceiling, a step function and a floor; Section 4.3)
F2	Parallel yields $\alpha < 1$	Parallel achieves $\alpha \geq 1$	Mixed
F3	Structured asymmetry required	Crystal forms without disorder	Consistent (in the source's model the active region vanishes for particles of the same size, whose interactions are reciprocal; a correspondence of structure that the source's own model already gives, so a consistency check, not a test)
F4	Five properties co-occur in recursive systems	Systems with four but not five	Mixed

Criterion	Prediction	Falsified if	Status
F5	ARC Bound in this paper's own form: a same-class sequential system does not sustain $\alpha > 2$ (decides nothing registered; Section 6.4)	Reproducible $\alpha > 2.3$, its 95% CI clear of 2.0, obtained on a system that alters its own weights, tools or artefacts from one round to the next and is corrected by mechanisms of its own class, that class declared and evidenced before the run, and obtained while correction still out-scales drift ($\beta_C > k$, the lower bound on $\beta_C - k$ sitting above zero) over a window pinned before the run began. Size on its own cannot overturn a stability limit: drop the co-scaling conjunct ($\beta_C > k$) and this test would fire on the very transient supercritical excursion the framework itself predicts. The three-outcome rule governs: any interval broader than the margin fixed before the run is recorded as INSUFFICIENT PRECISION, and counts as neither support nor refutation	Open (no interpretable exponent above one has been measured: Paper II 2.18's one interpretable fit is below one, and it finds Grok 4.1 Fast near ceiling); the registered bound is refuted only where P6 is refuted
F6	β_L determines α via $1/(1-\beta_L)$, read over a finite window as the complete solution of Axiom 2 gives it	The exponent the independently measured β_L implies disagrees with the fitted exponent beyond the margin of 0.10 that P5 registers	Untested
F7	Crossover R^* exists	Separately measured α and a place R^* inside the measured window and no linear→super-linear transition appears there (a pure power from the first pass is the case $a = \alpha$ and refutes nothing)	Untested
F8	Sequential requires output→input feedback	Parallel + shared state achieves $\alpha > 1$	Untested
F9	The acoustic time crystal's onset fits bounded composition, a growing mode that settles to a steady oscillation (this paper's own criterion: the time crystal is a consistency check and a cross-domain analogy, not a test of the framework, and no study of it is registered)	An onset recording that never settles, or a steady oscillation reached with no growing mode; the established account of the instability gives the same onset form (an exponential onset that settles to a steady oscillation), so agreement cannot tell the two accounts apart	Untested (every published recording is of a steady state); decides nothing registered
F10	$\oplus$ determines functional form	Measured $\oplus$ fails to predict scaling	Untested
F11	Classical growth-phase scaling ($\alpha \to 2$), this paper's own early-framework conjecture, outside the registration's scope	Growth-phase $\alpha \leq 1$ or $\alpha > 2.3$ in classical systems	Untested; decides nothing registered

Criterion	Prediction	Falsified if	Status
F12	The dimensional formula in biology, in the metabolic sense	Leaf venation's metabolic-scaling exponent deviates significantly from $2/3$	Untested
F13	External alignment yields $\alpha_{\text{align}} \approx 0$ (the external half of Section 6.5's conditional, registered as P19; the in-loop half, $\alpha_{\text{align}} \approx \alpha$, remains untested)	Architecture-dependent α_{align} with significant positive values	Not decided: the run of March 2026 cannot decide F13, its statistics being rank correlations and contrasts, not exponents, and its three flat and negative runs being the three the October 2026 code review found defective (Paper IV.a; Paper III 1.17)

We welcome falsification. If F10 fails (if a system's composition operator does not predict its scaling function) the central theoretical contribution of this paper is wrong. If F5 fails, this paper's own form of the ARC Bound fails; the registered bound is refuted only where P6 is refuted (Section 6.4). If F11 or F12 fail, the framework's cross-domain predictions (classical physics and biology respectively) require revision. Either outcome advances understanding.

Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge: github.com/MichaelDariusEastwood/arc-scaling-challenge.

8 What this paper does and does not show

What this paper shows is mathematical and holds inside its stated model: under the axioms and each case's hypotheses, the classification (Theorem 1), the exponent relation (Theorem 2), the necessity of the multiplicative interaction (Theorem 2a) and the crossover (Theorem 3); and the five properties on the premises Theorem 4 names. What it does not show is that any real system satisfies the axioms: the β_L-to-α relation, the crossover and the classification are untested in physical systems, the AI exponent's best fit is one model's and inconclusive, no exponent has been measured in a time crystal, and the ARC Bound, registered as a ceiling of at most two for same-class correctors, is untested, its point value two a conjecture. The table below sets what the paper does not claim beside what it does.

The empirical standing. The runs of March 2026 are precursor evidence that decides no registered proposition (Section 4.3): the sequential advantage was observed only where both exponents could be estimated, and Gu et al. (2026) report the opposite direction; the one interpretable capability fit is inconclusive and below one; and the alignment run cannot decide criterion F13 (Section 6.5). The mathematical framework is unaffected by them. What remains open is whether any real system satisfies its axioms, and the registered and drafted trials of the status paragraph below are the measurements that would say.

Table 10. What this paper does not claim.

Non-claim	Actual claim
That $\Lambda = 2.14$ and $\alpha \approx 2$ are 'the same number'	That different domains exhibit different functional forms determined by their composition operator. The AI exponent's best fit is one model's and inconclusive (Section 4.3), with both sequential point estimates below one, not ≈ 2
That this is proven science	That this is a testable hypothesis with thirteen specific falsification criteria
That neuroscience confirms the framework	That recurrence is structurally consistent but unquantified
Priority over experimental results	The experiments belong to Google, DeepSeek, NYU, COGITATE; the structural interpretation is ours

Category of contribution. The ARC framework is not an equation within an existing paradigm. It is closer to a cross-domain organising principle: a structural constraint on behaviour across substrates rather than an equation within one paradigm. The label identifies the *type* of contribution, not its evidential standing: organising principles of that type rest on generations of validation, and this one rests on preliminary evidence and thirteen falsification criteria.

Status, as at 31 October 2026. What this programme names laws are conjectures. Laws I to III were registered on 8 September 2026 and Laws IV and V on 15 October 2026 (version 1.103), the mathematics of Laws IV and V still in development (version 1.2, 29 October 2026). Each is a law under test: Law IV has not yet been tested, and the counted test of Law V's placement limb is Unit 83, below. The theory registration of twenty-two propositions was registered on 8 September 2026 at version 1.100 and updated on 13 September 2026 (version 1.102), 15 October 2026 (version 1.103) and 29 October 2026 (version 1.104, which adds no prediction and changes no law's statement); its terms are defined in a separate register. The first registered study, Unit 06, a classification audit that tests no law, was registered on 1 October 2026 and ran on 6 October 2026; one of its two runs could not start, so it is reported in full as administratively incomplete, with no inferential verdict. A second, Unit 83, the counted test of P21 under Law V, was registered on OSF on 22 October 2026 and ran on 27 and 28 October 2026: P21 is supported under the registered test, by the programme's own run, on the access axis, family-dependent, with its mechanism axis not evaluable and one call retried under a disclosed recovery amendment (results: DOI 10.17605/OSF.IO/98HKR). Two different axes are named here. The access axis is what Unit 83 varied: both reviewers are called each round, but the corrector sees the earlier kept entries with a marked new arrival and may revise it, while the monitor sees its accumulated register with the new arrival unmarked and can only accept or reject it. The performance axis is what it measured, the outcome, as against the mechanism axis, which was not evaluable. Law V, the ARC Embedding Law, remains a law under test: its placement limb, P21, is supported on the performance axis only, with its mechanism axis not evaluable, which is not support for the information-asymmetry mechanism; its build-order limb, read on P12, is undecided, since P12 is untested. A third, Unit 82, the counted test of P17, the premise beneath Law III's value two (Section 6.4), was registered on OSF on 27 October 2026 and has not yet run. The trials that would decide this paper's own predictions are drafts awaiting the author's submission. Every quantity in the formal model is defined in Appendix D and given a measurement in Appendix E, some in working definitions not yet tested, and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

9 Limitations

Evidential limitations. The earlier estimate of about 2.2 is retracted, and the one interpretable fit of the v5 cross-architecture experiment, Gemini 3 Flash's, is what survived it, one model's fit and inconclusive, not a correction of the same measurement (Section 4.3); the five capability runs of March 2026 are precursor evidence that decides no registered proposition. No α has been measured in time crystals. The COGITATE connection is structural, not quantitative. The β_L-to-α relationship is an identity proved in Appendix A.2 and checked numerically ($R^2 = 1.00000000$), an internal consistency check and not a validation of the relation in any system; the empirical measurement of β_L in physical systems requires independent confirmation. The R^* crossover is untested empirically. The β_L boundary values in the phase diagram are approximate. Conjecture 5 ($\oplus$ transitions) is supported in three computational systems, beside a Bernoulli control (Appendix F.6), and untested in physical experiments. The five-property test (Theorem 4) did not reach statistical significance ($p = 0.095$) and requires larger samples. The alignment run of March 2026, a single unregistered run, cannot decide criterion F13: its statistics are rank correlations and contrasts, not exponents, and its three flat and negative runs are the three the October 2026 code review found defective (Section 6.5).

Blind prediction testing. A blind prediction test was conducted on three computational systems (Barabási-Albert networks, gradient descent with momentum, Kuramoto oscillators). The measured α values were 3 to 20× smaller than predicted by $\alpha = 1/(1 - \beta_L)$. However, forensic analysis identified two independent confounds: (1) the numerical-derivative β_L estimation method is fatally biased, giving $\beta_L \approx 0.95$ regardless of true β_L, even for pure Bernoulli systems with 3 per cent Gaussian noise added; (2) none of the tested systems satisfy Axiom 2 (constant coupling coefficient a), and the BA network's effective coupling decreases $\sim 50\times$ over the simulation. Applied to systems built to satisfy Axiom 2, the linearisation method recovers the relation with $R^2 = 0.9999$, which checks the method, not the prediction for the three systems tested. The blind test failed for reasons found after it ran, so it neither supports nor refutes the relation; a new blind test with the method fixed in advance is required, and identifying natural systems that satisfy the axioms remains the central empirical challenge. Full methodology and forensic analysis are available as supplementary material (see Paper III).

Theoretical limitations. The framework formalises the *pattern* of recursive amplification but does not identify the *mechanism*. Theorem 2 is the solution of the growth equation Axiom 2 assumes, and Theorem 2a follows from the product form Axiom 1 assumes; their weight lies in what they make measurable, and the axioms themselves are untested in any physical system. It tells you *what* happens (scaling exponents, crossover depths, functional forms) but not *why* at a microscopic level (analogous to thermodynamics before statistical mechanics). Connecting the composition operator to microscopic dynamics remains an open problem.

10 Conclusion

Every system discussed in this paper (silicon neural networks, superconducting qubits, millimetre polymer beads, cortical neural circuits) maintains a low-entropy state far from equilibrium and, on this paper's reading, feeds its outputs back into its inputs. The framework predicts that below a critical depth the gains are ordinary and above it they compound, with the transition universal in structure and domain-specific in form; no such depth has yet been measured in any of them (criterion F7), and a claim that the transition is sharp would need a criterion of its own.

We have formalised this observation. The composition operator $\oplus$ sorts recursive systems by the functional equation their scaling satisfies: under the stated hypotheses it determines the functional form of scaling from measurable algebraic properties. The five derived properties (threshold, depth-dependence, base-quality dependence, multiplicative interaction, regime boundaries) follow as theorems from three axioms and, for two of them, the premises Theorem 4 names, not as empirical generalisations.

The framework issues predictions that the programme has been unable to locate in any theory confined to a single domain. It predicts the scaling behaviour of *new* recursive systems from their composition operator alone. It predicts a scaling crossover at a derived depth R^* that is independent of base quality and consistent across input qualities for a given architecture. It predicts that a measured β_L determines α via $\alpha = 1/(1 - \beta_L)$, a relationship proved for the axiom's solutions and checked numerically to $R^2 = 1.00000000$; the blind test of Section 9 neither supports nor refutes it for other systems. It states the ARC Bound as the theory registration holds it, a ceiling of at most two for correctors of the system's own class, which needs P6 and P20 and nothing else, with the point value two a conjecture. Its own earlier form, the coupling at which the sustained exponent reaches two ($\beta_L = 0.5$, $\alpha = 2$), is an early-framework conjecture that decides nothing registered (criteria F5 and F11), with the $O(N^2)$ tally of self-attention pathways only a heuristic for the frozen regime; on that conjecture two is a scaling limit, not a speed limit, which systems can exceed in transient supercritical excursions without staying stably self-correcting. And it conjectures that bounded systems exhibit composition operator transitions that could explain why different studies of the same system can observe different scaling regimes (Conjecture 5). These predictions and conjectures are specific, falsifiable, and forward-looking.

If the predictions fail, we will have learned where the structural analogy breaks down. If they hold, we will have identified a principle that connects intelligence, computation, and the physics of order across substrates.

Intelligence may not be a property of particular materials. It may be what happens on the far side of the recursive threshold.

Conjectural extrapolation. The furthest reading of the framework comes last, as a conclusion the mathematics does not deny rather than a result: Law I, a law under test, taken to the limit of capability does not rule out a universe or its creation. No measurement in this paper bears on that reading, no prediction of it is registered, and no test of it is in view, so it is graded as conjecture and is not established. It belongs to the long-horizon interpretive layer of *Infinite Architects* (Eastwood 2026a) and requires independent argument and evidence beyond what is reported here.

The predictions are specified. The falsification criteria are public. The data will decide.

Code and data. The paper's figures and results files are in the folder `papers/Foundational` of the public repository github.com/MichaelDariusEastwood/arc-principle-validation, and the scripts and results of the computational checks of Appendix F are in its folders `experiments/domain-validation__Foundational-and-Origin` and `experiments/blind-prediction-test__Paper-III-and-Foundational`, cited at commit `427b897d`. Two of the deposited results files in the paper's folder, `results_arc_1d_prediction_test.txt` and `results_complete_suite.txt`, credit the saprotrophic fungal rows to a study of 1975 and give a temperature of 25 degrees Celsius; the source Section 6.4 reads, Aguilar-Trigueros et al. (2017), credits those rows in its Table 1 to a study of 2013 and gives 28 degrees, and the files are kept as deposited. The paper and its earlier versions are deposited at doi.org/10.17605/OSF.IO/Y7QGD.

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Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.

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A Proofs

A.1 Proof of Theorem 1 (classification)

Case 1: multiplicative composition.

Proof. Hypothesis: $g : \mathbb{R}^+ \to \mathbb{R}^+$ continuous with $g(1) = 1$ and $g(R_1 \cdot R_2) = g(R_1) \cdot g(R_2)$.

Define $h(x) = \ln g(e^x)$. Then:

$$h(x + y) = \ln g(e^{x+y}) = \ln g(e^x \cdot e^y) = \ln[g(e^x) \cdot g(e^y)] = h(x) + h(y) \tag{A.1}$$

This is Cauchy's additive functional equation. Under the continuity of g (hence h), the unique solution is $h(x) = \alpha x$ for some constant α. Therefore $\ln g(e^x) = \alpha x$, giving $g(R) = R^\alpha$. □

Case 2: additive composition.

Proof. Hypothesis: $f : [0, \infty) \to \mathbb{R}^+$ continuous with $f(0) = 1$ and $f(r_1 + r_2) = f(r_1) \cdot f(r_2)$, in the added passes $r = R - 1$.

Taking logarithms: $\ln f(r_1 + r_2) = \ln f(r_1) + \ln f(r_2)$. This is again Cauchy's additive equation, with unique continuous solution $\ln f(r) = \alpha r$, giving $f(r) = e^{\alpha r}$, so that $g(R) = f(R - 1) = e^{\alpha(R-1)}$ and $g(1) = 1$, as Axiom 1 requires. □

Case 3: bounded composition.

Proof. $Q_{r+1} = \min(Q_r + \delta Q_r, Q_{\max})$. For any monotonically increasing sequence $\{Q_r\}$ bounded above by $Q_{\max}$, $\lim_{r\to\infty} Q_r \le Q_{\max}$. Therefore $f(R) = Q(R)/I$ is bounded and converges to a limit no greater than $Q_{\max}/I$: U saturates, at the ceiling only if the increments sum to at least the remaining gap, equality sufficing for attainment in the limit. □

A.2 Proof of Theorem 2 (scaling exponent)

Proof. From Axiom 2: $dg/dr = ag^{\beta_L}$, $\beta_L \in [0, 1)$, $g(0) = 1$, with $r = R - 1$ the passes after the first. Separating variables:

$$\int_1^g s^{-\beta_L}\, ds = \int_0^r a\, dr \tag{A.2}$$

$$\frac{g^{1-\beta_L} - 1}{1 - \beta_L} = ar \tag{A.3}$$

$$g(r) = [1 + (1 - \beta_L)ar]^{1/(1-\beta_L)} \tag{A.4}$$

For $R \gg R^*$ (where $(1 - \beta_L)aR \gg 1$):

$$g(R) \approx [(1 - \beta_L)aR]^{1/(1-\beta_L)} \propto R^{1/(1-\beta_L)} \tag{A.5}$$

Therefore $\alpha = 1/(1 - \beta_L)$. □

A.3 Proof of Theorem 3 (crossover depth)

Proof. The full solution $g(r) = [1 + (1 - \beta_L)ar]^{1/(1-\beta_L)}$, in the clock Axiom 2 declares ($r = R - 1$), can be analysed by defining $\rho = (1 - \beta_L)ar$:

$$g(r) = (1 + \rho)^{1/(1-\beta_L)} \tag{A.6}$$

For $\rho \ll 1$ (small r): $g \approx 1 + \rho/(1 - \beta_L) = 1 + ar$, approximately linear in r.

For $\rho \gg 1$ (large R): $g \approx \rho^{1/(1-\beta_L)} \propto R^{\alpha}$, power law.

The crossover occurs at $\rho \approx 1$, giving:

$$R^* = \frac{1}{(1 - \beta_L)a} = \frac{\alpha}{a}, \tag{A.7}$$

counted in added passes; counted from the first pass, the same depth is $1 + \alpha/a$.

R^* depends only on α and a, both independently measurable. It is independent of base potential I, consistent with the separability of Axiom 1. □

A.4 Why Axiom 2 takes this form

Why $dg/dr = ag^{\beta_L}$ and not some other functional form? This is the Bernoulli-type ODE for cumulative advantage (Price 1976), known in network science as preferential attachment, applied to the amplification factor g rather than absolute capability. The exponent β_L parameterises the degree of self-reference: $\beta_L = 0$ gives constant growth (no feedback); $\beta_L \to 1$ gives explosive growth (maximal feedback). The choice $\beta_L \in [0, 1)$ is the minimal model for a system where accumulated amplification accelerates future gains without divergence at finite R. More complex models (e.g., $dg/dr = ag^{\beta_L} - bg^{\nu}$, with ν a damping exponent) would introduce additional parameters; we adopt the minimal form and test it against observation.

B Version history and corrections

B.1 Version 5.21

Version 5.21, 31 October 2026. This version brings the paper into line with the programme as it stands on its date, after a line-by-line audit of version 5.20 against the theory registration, its signed records and the corrections register. No theorem's result, axiom or proof changes except as listed, and no date of any claim moves. (i) Unit 83's registered result is stated in full once, in Section 8, with its results DOI, where version 5.20 said that its report "is being sealed and checked, and will be published with its data", words already untrue when 5.20 was deposited; the first-page footnote and Appendix G point there, and Appendix C dates the run and the results. (ii) The ARC Bound is stated as the theory registration states it: at most two, for correctors of the system's own class, needing P6 and P20 and nothing else, with the point value two a conjecture. Version 5.20 printed it as $\alpha_{\max} = 2$ for "any classical sequential system", said that no biological network or economic process could sustain $\alpha > 2$, and said the value two "follows from one assumption"; the reach to biology and economics is withdrawn, the two assumptions behind the value are named, and this paper's own form of the bound is kept as an early-framework conjecture, tested by criterion F5, which with F11 decides nothing registered. The studies said to try the independence assumption were "AG and K"; the study that tries it is AH, and AG tests the ordering of the two derivations, which agree wherever $\beta_L = \gamma$ and not only at one point. (iii) Law II is stated as registered, its vanishing misalignment fraction given to Paper X's minimal model with positive gain drift, and Law III with the registration's conditions. (iv) The record of March 2026 is graded as its sources now

grade it: the five capability runs, dated from their own records, are precursor evidence that decides no registered proposition, with one interpretable fit, inconclusive, and the sequential advantage observed only where both exponents could be estimated; version 5.20 also printed a range of exponents from about zero to about 3, two named super-linear estimates and a coupling of about minus one read off the 0.49 fit, which are withdrawn, since $1/(1-\beta_L)$ is the asymptotic exponent. The alignment run cannot decide criterion F13, which version 5.20 graded "Refuted in original form"; Table 8 now gives the rank correlations it holds, named as such. The dated notes that kept the earlier gradings beside the corrected text move whole into Appendix B.5. (v) Criterion F9 stays, as this paper's own criterion, graded as a consistency check and a cross-domain analogy, not a test of the framework, with its rival account beside it; version 5.20 printed it as "Time crystal shows $\alpha > 1$ (in the transient growth regime, before saturation onset; the clause dates from version 5.7, 16 August 2026, and the row read 'in scaling regime' before it)", falsified by "$\alpha \leq 1$ in time crystal". The time crystal and the quantum rows of Table 4 are graded as analogies. (vi) Paper VII's 19 of 25 is given under its rule of March 2026, with its primary statistic and the dating its record supports, where 5.20 said its predictions "were dated before any fitting"; the twelve-domain extension gives 8 of 10 beside 9 of 11; Unit 56 is recorded as retired in favour of Unit 57. (vii) The opening states the author's why and the programme's stance on control, and Section 1 gives the programme in one paragraph; the ARC Principle names Law I as registered, the five-property conjunction being this paper's framework claim; usefulness stands beside the safety ratio; the values the loops carry are for international consensus; the cosmological close is stated as what the mathematics does not deny. (viii) Notation: the coupling is written β_L throughout the current text, as the notation register writes it (dated records keep the bare letter); Theorem 1's case (2) is solved in the clock Axiom 2 declares; the damping exponent is ν; γ has one name, the correction-leverage exponent; the second senses of d, p and k are listed. (ix) The two-dimensional fragmentation match is a reference value without a named source, not an analytic identity as version 5.20's entry, item (vi), described it. (x) References: Unit 83's results package and the founding paper of Recursive Dynamics are added; Paper III is cited as its OSF record now reads, at version 1.17; the Laws IV and V report's first publication is dated 1 October 2026; the pointer to the Eden Vision paper's comparison with prior work is withdrawn with its entry, Section 2 standing in its place. (xi) On release, on 31 October 2026, it also cites the registration's current version, 1.104, and the mathematics of Laws IV and V at version 1.2, and adds Unit 83's results reference. What version 5.20 printed in each changed place is kept in the deposited version 5.20.

B.2 Version 5.20

Version 5.20, 29 October 2026. This edition sets the paper in LaTeX to the programme's presentation standard: the front matter of version 5.19 moves into the sections and appendices Table 12 lists, and the sections are renumbered from 1. The third rung of the claim-status ladder, on cosmological and metaphysical conjecture, now stands at the close of the paper, and the note on the earlier senses of the letter U stands in Appendix D. The title is now the paper's question, and the OSF record keeps its title. The corrections made in this version are these. (i) Corrections 14 to 19, recorded at version 5.16, are carried into the statements, proofs and sentences they concern: Theorem 1 names each case's premises and states case (3) as a bound with its attainment condition; the solution of Axiom 2 and the crossover are written in the clock the axiom declares, in the proof of Theorem 2, in Theorem 3, beneath Theorem 2a and in Appendix A; Theorem 4 names the premises of its properties, states convexity at every depth and the exact doubling for the pure-power case; and the summary of results and the conclusion say that two of the five properties rest on added premises, that the crossover is independent of base quality, and no longer call the transition sharp. (ii) The acoustic time crystal is described as its source states it: a linearly stable limit cycle, whose coherence time of (100 ± 10) s, about 6,700 cycles, in one experiment, is not a ceiling, a saturation or a time taken to reach the cycle. Earlier versions called it saturating dynamics with about

6,700 cycles of coherence, coherence maintained before reaching a limit-cycle attractor, and a physical ceiling at which energy dissipation prevents indefinite growth. The source's quantity $\Delta_{ji}(n)$, a difference of powers of the two beads' sizes, is no longer read as a state difference growing over cycles; the reading of identical beads as a maximum-entropy state is replaced by the source's own finding; the exceptional point is described as computed; and criterion F3's status reads consistent where it read confirmed. (iii) The abstract, Section 1.1 and the caption of Figure 1 state what each cited programme measured, where earlier versions said that at least four independent research programmes had discovered capability gains exceeding linear accumulation, that DeepSeek-AI had shown parallel sampling to give diminishing returns, and that COGITATE had found recurrent processing sustaining representations, a finding that supported the framework's prediction. (iv) Law II is stated as version 1.103 of the theory registration states it, where earlier versions said that self-improvement holds together only while correction out-scales the drift it corrects; Laws IV and V are stated as registered on 15 October 2026, their mathematics cited at version 1.1; and the status line no longer says that Laws I to III are under registered adversarial test. (v) The book's names for the Eden Protocol's loops appear only as the book's. (vi) Physics is stated more exactly: the error suppression factor of quantum error correction applies for each increase of two in code distance; the Eddington limit is a luminosity ceiling, not a hard ceiling on accretion; Landauer's principle bounds the heat of erasing a bit and does not convert energy processed into bits; the unsourced measured exponent of about 1.1 for nonlinear gravitational collapse is withdrawn; the physics matches of the dimensional formula and the Friedmann matter era are stated as identities, and the percolation match as one made on an absolute value; the leaf-venation prediction states both senses of its exponent; and criterion F9 dates its growth-regime clause; and Figure 3 no longer plots a two-dimensional biological point that no measurement gives, nor prints the biology mean error the earlier figure marked as pending recompute, and its caption no longer says that one equation with zero adjustable parameters governs systems spanning biology, geophysics and condensed matter physics. (vii) References: the companion paper on the dimensional ladder is cited as On the Origin of Scaling Laws, where earlier versions linked the book; Paper IV is cited as Paper IV.a, which reports the alignment hierarchy; the book is linked to its deposit; the DeepSeek-AI paper is cited at its journal version, whose words are quoted; the Google Quantum AI entry notes the publisher's Author Correction of 28 April 2026, which relabels one figure and changes no figure quoted here; Cauchy's *Cours d'analyse* is linked to a digitised copy whose identity was confirmed; and the copyright line carries the author's standard wording. (viii) Wording: the principle, the bound and the protocol are introduced as argued, conjectured and set out. (ix) Status and claim level: the alignment tiers of the March 2026 run carry the caveat of the October 2026 code review and are no longer described as blind; p-values are no longer glossed as the chance of coincidence; the conclusion calls the ARC Bound a conjecture resting on a derived relation, where it said a derived structural constraint; and the Eden Protocol pilot is said to report no usefulness measure. (x) The count of the March 2026 run follows Paper II's current record, version 2.18: five frontier models were measured on both arms, capability and alignment, since Claude Opus 4.6 carried the alignment arm only. Where earlier versions counted six models in the capability arm, or stated the sequential advantage as universal across six models, this version gives that count and states the advantage as holding wherever both exponents could be estimated; the dated updates of March and September 2026 keep their own count, with Paper II's beside it. (xi) After the edition's adversary read, register corrections and source readings were carried in. Law II's criterion is no longer called stability, since it decides whether the misalignment fraction vanishes; the ARC Bound is called conjectured where it was called predicted, its basis is Law III's pacing derivation throughout, and the attention count is a heuristic (corrections 2, 4 and 9); the human-in-the-loop prediction is a programme hypothesis beyond the model; the comparisons with Engels and colleagues and with Liu and Meng follow Paper X; the parallel exponent of 0.31 is no longer called compute-matched; the sequential advantage carries Paper II 2.18's grading, observed, beside each dated update that called it confirmed or universal, and criterion F1 reads observed where it read confirmed; the sentence that the super-linear

prediction was never about frozen models is withdrawn; the work of Snell and colleagues, DeepSeek-AI, Zeng and colleagues, White and Seymour, and Kozłowski and Konarzewski is described as its sources state it; and the forms the axioms select are conceded as classical, as Paper X's novelty ledger records, with the question of an intelligence explosion dated to Good (1966). (xii) In the same pass: Theorem 5 of earlier versions, an empirical claim not derived from the axioms, as its own epistemic status says, is set as Conjecture 5, its words unchanged; the empirical sentences of Theorem 4's statement stand, unchanged, in a remark after it; the stability theorem is stated for the additive equation as Hyers (1941) proved it and for the exponential and multiplicative equations as the superstability of Baker et al. (1979), where earlier versions gave one bound for the multiplicative case; the surface code is no longer described as a layered, recursive code; the time crystal is no longer said to rise toward a ceiling set by an energy budget, and the four programmes are no longer said to show power-law scaling in AI or saturation in classical physics, nor to share no citations, funding or personnel; the cosmological reading of structure formation is stated as an analogy, and the Completeness Theorem as a statement that follows from Theorem 1; the version 3.1 notes no longer read the frozen regime as weak coupling or as the boundary of the power-law regime, since a coupling near minus one lies outside the model's range; the physics instances of the dimensional formula are given as reference values, two of them without a named source, and the derivations of the formula are no longer counted as at least seven; and the claims, glosses and conclusion are held to what the model and the record show, Appendices E and F saying what the deposited code holds. (xiii) Before release, on 29 October 2026, the status is brought to that date: Section 8 and Appendices C and G give the theory registration's version 1.104, approved that day, Unit 83's registration and run, and Unit 82's registration; Section 6.4 names the component of the independence-accumulation assumption that Unit 82 tests; and, as version 1.104 reads them, the runs of March 2026 are precursor evidence, with arms not matched in compute. The title block names the paper's area, Theory, and the first-page footnote links the book and the page about the author. (xiv) The fungal value of Table 6, Figure 2 and the paragraph between them is given as its source reports it, the mean of three published colony-level factors, where version 5.19 printed a standard error of 0.07 beside it, which is one factor's standard error, not the mean's. No theorem's result, prediction's number or criterion's kill condition changes except as listed, and no date of any claim moves.

B.3 Corrections 1 to 20, as printed

The author's note on mathematical corrections, as version 5.19 printed it at the head of the paper, is kept whole below; its section and appendix numbers are those of the earlier layout, which Table 12 maps to this edition. The note's first paragraph says that these papers have not undergone formal human peer review; what this paper has been through is internal and never independent: adversarial refutation rounds, automated gates run on every release, source and quotation checks against each cited work, and dated revisions with the corrections this appendix lists, a system shown whole at michaeldariuseastwood.com/verify. The statement on tool assistance it points to now stands in the Acknowledgements.

Author's Note: Mathematical Corrections

This version incorporates the following corrections to the broader framework. These papers have not undergone formal human peer review; the author's statement on tool assistance is in the declaration at the end of this paper.

1. Axiom 2 Correction: The Cumulative Advantage ODE now operates on the amplification factor $g(R)$ with $g(0) = 1$, matching Paper III. Previous versions applied the ODE to accumulated capability Q with $Q(0) = I$, which coupled the base potential I into the recursive dynamics and violated the separability

asserted in Axiom 1 ($U = I \times g(R)$). The corrected formulation yields the simpler crossover formula $R^* = \alpha/a$, independent of I.

2. Quadratic Limit Regrounding: The derivation of $\alpha \leq 2$ from stability analysis, elasticity, and Lyapunov/edge-of-chaos arguments has been **withdrawn** as these arguments contained category errors (elasticity > 1 does not imply dynamical instability; an autonomous flow on the line is monotone, so it cannot so much as oscillate, still less turn chaotic, and in two dimensions the Poincaré-Bendixson theorem shuts chaos out as well). The corrected justification is information-theoretic: transformer self-attention creates at most $O(N^2)$ pairwise information pathways, imposing a quadratic ceiling on information density per recursive step. Counting pathways this way is a heuristic and proves nothing about capability, and the move from quadratic pathway density to $\beta = 1/2$ is a further conjecture, as is the supposition that any of it carries across to classical sequential systems at large. What now supports the value two in the theory is the pacing derivation of Law III at $\gamma = 1/2$, given in correction 9 below and in the statement paper.

3. Kleiber Reframing: Kleiber's Law ($M^{0.75}$, implying $\alpha \approx 1.33$) is now explained through thermodynamic drag, the physical costs of pumping fluids through three-dimensional fractal networks, dissipating heat, and distributing resources against gravity, rather than as a geometric constraint preventing a stability-limited maximum. The formula $\alpha = d/(d + 1)$ itself has been independently derived by multiple groups (West et al. 1997, 2004; Banavar et al. 1999; Demetrius 2008; Zhao 2022; Bettencourt 2013); the ARC framework's contribution is identifying Cauchy-constrained recursive composition as the reason these independent derivations converge. Demetrius and Tuszynski (2010) independently derived the same formula from quantum oscillator coupling, producing identical predictions.

4. Epistemic Status: The equation $U_{\max} = I \times R^2$ is offered as a conjecture rather than a result, and the limit it sets on classical sequential computation is relative to architecture: an architecture-relative capacity ceiling of the kind information theory supplies, not a physical stability law.

These refinements preserve all core predictions: the β-to-α derivation, the five structural properties, the composition operator classification, and the falsification criteria remain unchanged (a thirteenth criterion, F13, on alignment scaling, is introduced in Paper III).

5. v3.0 Additions: This version adds the Six Hidden Assumptions of Cauchy's classification (§2.2), the Hyers-Ulam stability theorem (Hyers 1941; Ulam 1960) establishing that each of the recognised scaling forms is stable within its own equation (a function approximately satisfying one of Cauchy's equations lies near an exact solution of *that same* equation), the Attractor Theorem combining these results with the β-derivation, and a structural implication connecting the composition operator classification to the space of possible classical scaling laws (§4.4). The earlier framing of the three forms as "stable attractors in function space" is withdrawn: the stability is per-equation, not a partition of function space, because a function satisfying none of the equations is not drawn toward any of them.

6. v3.1 Additions (March 2026): This update integrates empirical results from the v5 cross-architecture experiment (Paper II), which tested 6 frontier AI models (Grok 4.1 Fast, Claude Opus 4.6, GPT-5.4, DeepSeek V3.2, Gemini 3 Flash, Groq Qwen3) on 30 problems with bootstrap confidence intervals. Key findings: (a) sequential advantage is universal ($\alpha_{\text{seq}} > \alpha_{\text{par}}$ for every model), (b) the best-fitting power law, on Gemini 3 Flash, is one model's fit and inconclusive, its sequential point estimates positive but sub-linear (§3.3 prints its row whole), (c) the scaling exponent is architecture-dependent, ranging from near-zero to ~ 3, and (d) the quadratic bound conjecture ($\alpha \leq 2$) is neither confirmed nor refuted. Alignment scaling results from Paper IV are also integrated: α_{align} is architecture-dependent, ranging

from −0.25 (Gemini) to +0.44 (Claude Opus 4.6), with a three-tier hierarchy. These findings preserve the mathematical framework while significantly revising the empirical predictions for AI systems. Sections updated: §3, §4.1, §5, §6, §7, §8.

7. v4.0 Additions (March 2026): Brief note on the Eden Protocol's first in-programme empirical pilot (§5.5). No changes to the mathematical framework.

8. v5.0 Corrections (24 March 2026): Four corrections identified by review. (1) The d=2 biological prediction (jellyfish/cnidarians) is reclassified from 'confirmed' to 'unconfirmed' pending identification of a valid test organism with genuinely two-dimensional resource transport. The Friedmann cosmological confirmation at d=2 is retained. (2) The d=3 mammalian exponent is acknowledged as debated (0.67 to 0.75 range), with the framework predicting this variation rather than a single universal value. (3) The derivation of $\alpha = d/(d+1)$ is clarified to require three conditions (multiplicative composition, space-filling geometry, and a conservation or optimisation constraint), not space-filling alone. (4) Demetrius and Tuszynski (2010) added as an independent derivation from quantum oscillator coupling. No changes to the mathematical framework or core predictions.

9. Stability Ceiling Correction (16 August 2026): The ceiling was printed as $\alpha_{\text{crit}} = 1/\gamma$. Under this paper's own definition of γ that form runs the wrong way: it makes a better corrector lower the ceiling, and the same-class cap $\gamma \leq 1/2$ would give a ceiling of at least two rather than at most two. The corrected form is $\alpha_{\text{crit}} = 1/(1-\gamma)$, the reciprocal of the corrector's shortfall from full proportionality, obtained from the pacing condition $\alpha\gamma > \alpha - 1$. At $\gamma = 1/2$ both forms return two, which is why the error survived review; a discrimination study at values away from one half, where the two families separate, is drafted and awaits human submission. The superseded form is retracted and entered in the corrections log.

10. Physics Aggregate Withdrawn (14 August 2026): The previously reported below-0.2 per cent mean error across physics domains is withdrawn. A mean taken over analytic identities is not a measurement of predictive accuracy. The individual domain entries stand on their own terms; the aggregate does not, and it is not restored in weakened form.

11. Earthquake Datum Withdrawn (14 August 2026): The earthquake B-value row is a definitional identity rather than an independent measurement of the exponent, and is removed from the physics comparison and its figures. It is retained separately as a variable-pair case study, where it is informative about the method rather than about the prediction.

12. Hyers-Ulam Scope Narrowed (14 August 2026): The statement that the three scaling forms are stable attractors in function space is withdrawn. Hyers-Ulam stability is per-equation: a function approximately satisfying one of the functional equations lies near an exact solution of *that same* equation, and a function satisfying none of them is not drawn toward any. The formal stability statement is preserved; the ubiquity implication is separated as a distinct empirical question that Hyers-Ulam alone does not settle.

13. Body Reconciled With Corrections 10 and 11 (1 September 2026): Corrections 10 and 11 withdrew the earthquake B-value row and the sub-0.2 per cent mean, yet both survived in the section 5.4 physics table and in the sentence that summarised it. Neither appears in the body any longer: each of the four instances that remain is assessed on its own merits, and no aggregate figure is stated.

14. Theorem 1's Premises Separated (19 September 2026): The theorem's header binds all three cases to Axioms 1 to 3. Under this paper's own definitions that header is wider than the results need. Solved in the clock Axiom 2 declares, the axiom gives $g(R) = [1 + (a/\alpha)(R - 1)]^{\alpha}$ for $R \geq 1$. Case (1)'s pure power agrees with that solution only where $a = \alpha$; case (2)'s exponential would require $\beta = 1$, which the axiom's declared domain $\beta \in [0, 1)$ excludes; and case (3) never saturates under it, because the solution is unbounded for $a > 0$ and $\alpha > 0$. The corrected statement is that cases (1) and (2) rest on Axiom 1, Axiom 3 and each case's own composition hypothesis, exactly as Appendix A.1 proves them, with Axiom 2 not among their premises; and that case (3) rests on the capped recurrence it states, its ceiling remaining separable from base potential only if the ceiling is itself proportional to base potential. All three results stand as printed. The Cauchy classification needs no differential equation, and Theorem 2's relation $\alpha = 1/(1 - \beta)$ falls out of the case (1) check unchanged.

15. The Declared Clock Carried Into the Solution (19 September 2026): Axiom 2 declares the added-recursion clock $r = R - 1$, so that $g = 1$ at $r = 0$ restates Axiom 1's $g(1) = 1$. The solution is then printed with R standing in for r at Theorem 2, at Theorem 3 and twice in Appendix A, and Appendix A.2's integral in dr carries an upper limit of R. Evaluated at the declared baseline, the printed form returns $[1 + a/\alpha]^{\alpha}$, which exceeds one for every admissible rate constant where the axiom requires one. The corrected statement is that the solution in the declared clock is $g = [1 + (a/\alpha)r]^{\alpha}$ for $r \geq 0$, equivalently $g(R) = [1 + (a/\alpha)(R - 1)]^{\alpha}$ for $R \geq 1$, and that each equation should say which clock it counts in; the zero-depth reading beneath Theorem 2a is to be taken in the declared clock, where zero added passes is the first pass. The consequence for the crossover is a shift by one and nothing else: α/a is the crossover measured in added passes, and the same depth counted from the first pass is $1 + \alpha/a$. Theorem 2a is otherwise untouched: run with the declared clock's baseline throughout, its contradiction still stands, so the theorem is sound and its symbols alone were wrong. The repair made at Axiom 2 in v5.14 is real and is not reopened here; what this correction reaches is the places downstream of it.

16. Theorem 4's Properties and Their Premises (19 September 2026): The theorem states that all five properties follow from Axioms 1 to 3 alone, regardless of which composition operator applies, and the conclusion repeats that they follow as theorems and not as empirical generalisations. Two of the five carry more than the axioms. Property 4's exact doubling law, that doubling recursive depth multiplies capability by 2^{α} at any base potential, holds only where $a = \alpha$, the condition correction 14 records; under bounded composition it fails outright, because iterating it grows without bound against a finite ceiling, so the phrase regardless of which composition operator applies is false as printed. Property 5 is justified in its own sentence by finite energy and external error mechanisms, and neither appears anywhere in the three axioms. The corrected statement is that property 3 and the base-quality half of property 4 follow from Axiom 1 alone; that property 1 follows from Axioms 1 to 3 through Theorem 3, as the property's own justification states; that the exact 2^{α} doubling holds in the pure-power case and asymptotically otherwise; and that property 5 rests on a named added premise about finite resources and external error. What is preserved unweakened is the conjunction of the five as the framework's distinctive prediction, the observation that the conjunction is what is uncommon, and the falsification criterion that a system with four of the five but not the fifth would constrain the framework.

17. Convexity Holds at Every Depth (19 September 2026): Property 2 states that capability is strictly increasing and convex above the crossover, and justifies it by $g(R) = R^{\alpha}$ being convex for $\alpha > 1$. Differentiating Axiom 2 directly gives $d^2g/dr^2 = a^2\beta g^{2\beta-1}$, which is strictly positive at every depth for $a > 0$ and $\beta > 0$, and zero at $\beta = 0$. The printed sentence is therefore true and weaker than the truth, and the justification quoted for it reads the asymptotic pure power rather than the solution of the axiom. The

corrected statement is that amplification is convex at every depth once the coupling is positive, and that R^* marks the depth at which compounding becomes numerically dominant rather than the depth at which curvature begins. The crossover is sited by the convention that the dimensionless argument is near one, which Appendix C.4 already names as an approximation carrying a 2 to 7 per cent error in the absolute value, so a claim that the transition is sharp needs a criterion of its own. What is preserved is R^* as a derived quantity, a function of the scaling exponent and the rate constant alone, independent of base potential, and checkable by measuring those two separately.

18. Case 3 Proves a Bound, Not Attainment (19 September 2026): Appendix A.1 case 3 argues that a monotonically increasing sequence bounded above by the ceiling has a limit no greater than the ceiling, and concludes that capability saturates; Theorem 1 case (3) states the saturation value as the ceiling divided by base potential. The appendix proves the bound and not the value. A capped monotone process converges to a limit at or below the ceiling, and it reaches the ceiling only if the increments sum to at least the remaining gap, equality sufficing for attainment in the limit. The corrected statement is that the appendix establishes a limit no greater than the ceiling, and that the theorem's value needs either the attainment condition stated beside it or the limit written as a bound. Requiring the partial sums to exceed the gap would be too strict, since the equality case attains the ceiling asymptotically: one overstatement is not to be replaced by another. What is preserved is that a capped monotone process converges, which is true and proved, and saturation as a qualitative regime.

19. The Crossover Does Not Move With Base Quality (19 September 2026): The conclusion lists among the framework's predictions a scaling crossover at a derived depth that shifts systematically with base quality. Five places in this paper say the opposite: correction 1 above, section 2.4, the falsification criterion at section 5.3, Appendix A.3, and Appendix C.4, which records that varying base potential does not shift the crossover. The derivation settles it in either clock, because base potential appears nowhere in Axiom 2 and nowhere in its solution, so α/a and $1 + \alpha/a$ are both free of it. The provenance is on this page: correction 1 records that previous versions applied the equation to accumulated capability with base potential as the initial condition, which coupled base potential into the recursive dynamics, and under that superseded formulation a crossover moving with base quality was the correct prediction. The corrected statement is that the crossover is predicted to be independent of base quality and consistent across input qualities for a given architecture, matching Theorem 3, Appendix A.3, the falsification criterion and the computational validation. As with corrections 10 and 11 at correction 13, the defect is an earlier correction that did not reach the body. Every other prediction in that list stands unweakened: prediction of scaling behaviour from the composition operator alone, the coupling-to-exponent relation and its computational validation, the ARC Bound as a proposed stable-scaling limit with its architecture relativity and its expiring basis, and the composition operator transitions of Theorem 5.

20. The Laws, the Estimator Row and the Record Brought Up to Date (3 October 2026): Until this version, the line beneath the author block and the box on this paper's place among the laws said that the theory has three laws, the three ARC Laws, and the epistemic-status line at the end called every law the programme names a conjecture under registered adversarial test. The theory has had five laws since 24 September 2026: Laws I to III were registered on 8 September 2026, and Laws IV and V are neither registered as laws nor tested, their mathematics set out at version 1.0 of 3 October 2026. Several passages, correction 6 among them, printed Paper II's regression exponent or its coefficient of determination apart from the rest of its row, some calling the exponent the best cross-architecture estimate, and Theorem 4's second property printed the exponent beside the bootstrap interval that belongs to the endpoint estimate of the same row. The abstract read that fit as confirming positive power-law scaling, and §5.4 read it as

making the ARC Bound question moot for current systems. Each passage now carries the row whole, or names the fit as one model's and inconclusive and points to §3.3, where the row is printed whole. §5.4 named study AE among the studies on trial and said that it registers the reciprocity requirement; it has since been withdrawn from the programme. §4.1 and §5.4 called the test of the identity between the two derivations of the critical exponent registered, and §4.1 called the engineered cells registered; those designs are drafts preregistered by timestamp and not registered on OSF, and the passages now say so. §5.1 and §5.4 said of the replication of Paper VII's comparison that its registration was in preparation and not yet approved; it is preregistered by timestamp (units 55 and 56, first anchored on 12 August 2026) and not yet registered on OSF. In the summary table of §3.1 the cell for Google Willow carried no date; it now carries both of its dates and is graded convergent timing, not a prediction. Correction 8 now says that its four corrections were identified by review; the provenance of that review is not recorded, and the word that called it independent is withdrawn. A dated note in §3.3 says that the figures of the v5 table are the estimates of March 2026 and that Paper II's later Table 6b gives other values. The record of dated predictions of 25 August 2026 says that the registry submission click remains outstanding across the programme; it stands as recorded, and a dated note after it gives the state as read on 3 October 2026: the theory's registration was accepted on 8 September 2026, and one trial was submitted for registration on 1 October 2026. No theorem, result, figure or falsification criterion is withdrawn.

B.4 The notes of versions 5.15 to 5.19

The notes of versions 5.15 to 5.19, as printed, newest first.

v5.19 (5 October 2026): the title now follows the programme's uniform order (paper, its law or framework, its own title) and reads *Foundational Paper · The ARC Theory · Law I, the ARC Principle · Recursive Amplification as a Cross-Domain Structural Principle*; until this version it read *The Foundation of Law I, the ARC Principle: Recursive Amplification as a Cross-Domain Structural Principle*, and the title line of the footer, which read *THE ARC PRINCIPLE: FOUNDATIONAL PAPER*, now carries the title. The plate beneath the author line now gives the paper's place in the programme in the words of the programme's map. Mentions of the programme's papers in this paper now carry their uniform titles. The summary beneath the title now says that the programmes it describes are cited for convergence and not for mechanism, and that whether any of their systems accumulates is untested in each case. The note on how to read this paper now says that, under the paper's axioms, the composition operator determines the scaling form for the three composition rules the paper sets out, where it said that the operator uniquely determines the scaling form. Paper II is cited at its current version, 2.17. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v5.18 (4 October 2026): the title now names the law it founds, Law I, the ARC Principle; until this version it was titled *The ARC Principle: Recursive Amplification as a Cross-Domain Structural Principle*. The note on how to read this paper now says that it is the foundation of Law I and the formal core beneath Laws I to III, as the box beneath the author block says, where it had said that the paper carries a law. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v5.17 (3 October 2026): the paper is brought into line with the theory's five laws and with the record as it stands. Substantive: the line beneath the author block and the box on this paper's place among the laws now say that the theory has five laws, Laws I to III registered on 8 September 2026 and Laws IV and V added on 24 September 2026, neither registered as a law nor tested, with their mathematics at version 1.0

of 3 October 2026; a dated note in that box places this paper against Laws IV and V; the epistemic-status line gives Laws IV and V their status; every passage that printed a figure of Paper II's row for Gemini 3 Flash apart from the rest of the row, or its regression exponent beside the interval of the endpoint estimate, now carries the row whole or names the fit as one model's and inconclusive and points to §3.3, where the row is printed whole; study AE is recorded as withdrawn; §4.1 and §5.4 give the test of the identity and the engineered cells as preregistered by timestamp, where they said registered; the replication of Paper VII's comparison is given in §5.1 and §5.4 as preregistered by timestamp (units 55 and 56), not yet registered on OSF; the cell for Google Willow in the table of §3.1 carries both of its dates; a dated note in §3.3 dates the figures of the v5 table to March 2026 and names Paper II's later Table 6b; correction 8 reads identified by review; the record of dated predictions of 25 August 2026 stands as recorded, and a dated note after it gives the state of the registrations as read on 3 October 2026; correction 20 records what was printed before. Corrections to the references: entries 7, 12, 13, 20, 23 and 24 are corrected against their registries, and the first author of entry 7 is corrected to the registry's in the three places the text names the authors. Editorial: Paper II is cited at its current version, 2.15; the works the paper cites are linked to their official sources; and the version and date carriers move to v5.17. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v5.16 (21 September 2026): six dated corrections have been added to the author's note, numbered 14 to 19, and a scoping update has been added beneath the falsification status summary. Substantive: correction 14 separates Theorem 1's premises, with all three results standing as printed; correction 15 carries the added-recursion clock declared at Axiom 2 into the printed solution and into Appendix A; correction 16 names the added premises behind two of Theorem 4's five properties; correction 17 states that amplification is convex at every depth once the coupling is positive and that the crossover marks where compounding becomes numerically dominant; correction 18 states that Appendix A.1 case 3 establishes a bound rather than attainment, and gives the attainment condition; correction 19 records that the crossover is predicted to be independent of base quality, the conclusion's contrary clause being the residue of the formulation superseded at correction 1. Also substantive: the scoping update records that Paper II now frames the cross-architecture replication on five analysable models of the six tested. Editorial: Paper II is cited at its current version and revision date in the references. No theorem, result, figure or falsification criterion is withdrawn, no withdrawn material is restored or weakened into a replacement, and the first published date is unchanged.

v5.15: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.

B.5 The dated notes moved from the body in version 5.21

Version 5.21 moves here, as version 5.20 printed them, the dated notes that stood in the body beside text since corrected; the body now carries one current statement in their place. Each is headed by the place it stood in version 5.20; its references point to this edition's sections, and its figures and gradings are those of its date.

Section 4.2, property 1 (AI)

v3.1 note: The v5 data confirms the universal sequential advantage but shows that not all models achieve $\alpha > 1$; the threshold separates sequential from parallel modes rather than sub-linear from super-linear scaling. (Section 4.3 gives Paper II 2.18's current grading: observed.)

Section 4.3, the update of March 2026 with its table and its note of 3 October 2026

Update: v5 cross-architecture results (Paper II, March 2026). The v5 experiment tested 6 frontier models on 30 problems with bootstrap confidence intervals (in Paper II's current record, version 2.18, five were measured on both arms, capability and alignment, since Claude Opus 4.6 carried the alignment arm only). Key findings that bear on the evidence presented in this section:

Table 11. The v5 cross-architecture results of March 2026.

Model	α_{compute}	r^2	Sequential > parallel?
Grok 4.1 Fast	~ 3.0	Low	Yes
Claude Opus 4.6	~ 1.5	Low	Yes
GPT-5.4	~ 0.3	Low	Yes
DeepSeek V3.2	~ 0.2	Low	Yes
Gemini 3 Flash	0.49 (regression, standard error 0.20); endpoint 0.59, bootstrap 95 per cent interval minus 1.3 to 2.9; parallel 0.31; one model's fit and inconclusive	0.86	Yes
Groq Qwen3	~ 0.1	Low	Yes

Note (3 October 2026). *The figures in the table above are the estimates of March 2026, and Paper II's later Table 6b gives other values; only the row for Gemini 3 Flash is printed here with Paper II's current figures.*

Beside this note, the current record: Paper II's current record, version 2.18, states that five models carried the capability arm, Grok 4.1 Fast, DeepSeek R1, Gemini 3 Flash, GPT-5.4 and Groq Qwen3, that Claude Opus 4.6 was measured only in the alignment arm, and that sequential exceeded parallel wherever both exponents could be estimated, Grok 4.1 Fast giving no parallel exponent; the table's Claude row, its DeepSeek label and its 'Yes' for Grok 4.1 Fast are March 2026 entries that Paper II does not carry.

What is confirmed: (1) Sequential > parallel universally ($\alpha_{\text{seq}} > \alpha_{\text{par}}$ for every model). (2) Power-law form fits well for Gemini 3 Flash (its row in the table above, one model's fit and inconclusive). (3) Error reduction with compute investment is real.

What is NOT confirmed: (1) Super-linear scaling ($\alpha > 1$) is not universal across architectures; the best fit, Gemini 3 Flash's in the table above, is one model's fit and inconclusive, and both of its sequential point estimates are sub-linear. (2) No universal α value exists; it varies from ~ 0 to ~ 3 depending on model and problem set. (3) The quadratic bound conjecture is neither confirmed nor refuted.

The framework should now be read as: *'Sequential recursion yields $\alpha_{seq} > \alpha_{par}$ universally, with the magnitude of α being architecture-dependent. The key structural claim, that recursive composition produces stronger scaling than parallel composition, is confirmed. The quantitative claim that $\alpha > 1$ for all sequential systems requires revision.'*

Beside this update of March 2026, the current grading: Paper II's current record, version 2.18, grades the sequential advantage as observed, on the evidence "Wherever both exponents could be estimated: Gemini 3 Flash; DeepSeek R1, unreliably", with no parallel exponent for Grok 4.1 Fast, and Gu et al. (2026) report the opposite direction. Version 1.104 of the theory registration (29 October 2026) records that the two arms of these runs of March 2026 were not matched in compute, and reads the runs as precursor evidence: they came before the registration and decide no proposition it registers. Every statement in this paper that the advantage is universal or confirmed is superseded by that grading.

Section 5.1, the original prediction

ARC Bound prediction (original). Optimised large language models with explicit self-correction mechanisms will converge toward $\alpha \approx 2$ ($\beta \approx 0.5$). Systems with explicit critique-revise architecture will show $\beta \approx 0.45$ to 0.55 ($\alpha \approx 1.8$ to 2.2), while systems with simple chain-of-thought will show $\beta \approx 0.2$ to 0.3 ($\alpha \approx 1.25$ to 1.4). This is testable within current model generations.

Section 5.1, the version 3.1 empirical update

Version 3.1 empirical update. The v5 experiment (6 models, 30 problems; in Paper II's current record, version 2.18, five were measured on both arms, capability and alignment, since Claude Opus 4.6 carried the alignment arm only) does not support the original prediction. The best fit is Gemini 3 Flash's, for which Paper II reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of minus 1.3 to 2.9, a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a parallel exponent of 0.31 by endpoint estimate; it is one model's fit and inconclusive. Its regression exponent implies $\beta \approx -1.0$ under $\alpha = 1/(1-\beta)$ instead of $\beta \approx 0.5$. That figure sits outside the $[0, 1)$ range Axiom 2 allows, which places the frozen regime beyond what the Bernoulli model as stated covers rather than inside it as a weak-coupling case; Paper II names it the physical regime, $\beta < 0$. No deployed system measured to date has therefore been observed inside the regime Theorem 2 and Laws II and III, each a law under test, govern; the experimental cells of designs preregistered by timestamp are built to enter it, and the laws are predictions about that regime rather than descriptions of the one today's systems occupy. Sub-linear scaling ($\alpha < 1$) suggests that current AI models sit outside the coupling range the model covers ($\beta < 0$), where accumulated context does not accelerate later steps. Two models (Grok 4.1 Fast, $\alpha \approx 3$; Claude Opus 4.6, $\alpha \approx 1.5$) show super-linear point estimates, but with low r^2 values, indicating poor power-law fits (Paper II 2.18 measured Claude Opus 4.6 in the alignment arm only; Section 4.3). The prediction that optimised systems will converge toward $\alpha \approx 2$ remains open but is not supported by current data. The revised empirical picture is that α is architecture-dependent, and the primary validated claim is that sequential processing yields higher α than parallel processing for every model tested (Paper II 2.18 now grades this as observed, wherever both exponents could be estimated; Section 4.3).

Section 5.4, the version 3.1 note

v3.1 note: What the v5 data points to is that current AI systems may operate in a coupling regime weaker than first assumed ($\beta < 0$): the regression exponent of the best fit, one model's and inconclusive (Section 4.3 prints its row whole), yields $\beta \approx -1.0$ under the relation $\alpha = 1/(1-\beta)$, and the $[0, 1)$ range set by Axiom 2 does not admit such a value. This would place current AI systems in the diminishing-returns regime, outside the power-law regime, not deep within it as originally hypothesised.

Section 6.2, the version 3.1 note

v3.1 note: The v5 cross-architecture data shows that α varies significantly across models (~ 0 to ~ 3), consistent with different architectures having different effective β values. This is consistent with the regime transition prediction, though it does not confirm it directly since the variation is across architectures rather than within a single system.

Section 6.4, the retracted initial estimate and its update of March 2026

Initial estimate (retracted): $\alpha \approx 2.2$ ($n = 12$, 95% CI: 1.5 to 3.0). Note: this point estimate exceeded the conjectured ARC Bound of $\alpha \leq 2$; however, the 95% confidence interval [1.5, 3.0] was wide enough to be consistent with both the theory and its negation, and should not be treated as confirmatory of either bound violation or bound satisfaction. *Update:* Paper II (March 2026) extends this test to 6 frontier models with 30 problems and bootstrap confidence intervals (in its current record, version 2.18, five were measured on both arms, capability and alignment, since Claude Opus 4.6 carried the alignment arm only). The cross-architecture best fit is one model's, Gemini 3 Flash's, and inconclusive (Section 4.3 prints its row whole), and the bootstrap interval of its endpoint estimate contains the value two; both of its sequential point estimates, though, are well below 2, so for current systems the empirical challenge has shifted from 'can α exceed 2?' to 'can α reliably exceed 1?' Individual model point estimates range from ~ 0 (Groq Qwen3) to ~ 3 (Grok 4.1 Fast), but models with $\alpha > 1$ have low r^2, suggesting poor power-law fits rather than genuine super-linear scaling. The ARC Bound conjecture is neither confirmed nor refuted by the v5 data; it remains an open conjecture that will become empirically testable if and when AI systems consistently achieve $\alpha > 1$.

Section 7, the version 3.1 falsification status summary

Version 3.1 falsification status summary. The v5 experiment has updated the status of three criteria: **F1** (sequential advantage) is confirmed in its revised, weaker form ($\alpha_{\text{seq}} > \alpha_{\text{par}}$, rather than $\alpha > 1$); Paper II 2.18 now grades it as observed (Table 9). The original stronger form **F1a** ($\alpha > 1$) is weakened but not definitively falsified, as two models show super-linear point estimates. **F13** (alignment scaling) is refuted in its original form: α_{align} is not universally near zero but is architecture-dependent, with values ranging from -0.25 to $+0.44$ (a single unregistered run, under the caveat of Section 6.5). The falsification of F13 in its original form is an important correction: alignment scaling is a property of the architecture-training interaction, not solely of whether alignment is 'in the loop.' This finding motivates a revised F13 in future versions.

Section 7, the update of September 2026

Update: scope of the sequential advantage (Paper II, September 2026). Paper II was revised on 18 September 2026 and now frames the cross-architecture replication on five analysable models of the six tested. It scopes the sequential statement to every model in which both arms were measurable, and states the parallel result as at or near zero in every model measured, with one measured exception, Gemini 3 Flash. The regression estimate on that model, its coefficient of determination and the 18 of 30 problems that entered the exponent fit are unchanged. For Gemini 3 Flash, Paper II reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of minus 1.3 to 2.9, a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a parallel exponent of 0.31 by endpoint estimate. It is one model's fit and inconclusive. Where this paper states the sequential advantage as universal across six models, the denominator carrying it is the five in which both arms were measurable; six is the number tested. Its current record, version 2.18, counts five frontier models measured on both arms, capability and alignment, since Claude Opus 4.6 carried the alignment arm only; it grades the sequential advantage as observed wherever both exponents could be estimated (Gemini 3 Flash; DeepSeek R1, unreliably), and gives the parallel exponent as near zero in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3 (all 30 problems), and none for Grok 4.1 Fast.

Section 8, the update of March 2026

Update of March 2026, its words kept: revised empirical standing. The v5 cross-architecture experiment (Paper II, March 2026) has tested the AI predictions against six frontier models (in Paper II's current record, version 2.18, five were measured on both arms, capability and alignment, since Claude Opus 4.6 carried the alignment arm only). The results require an honest reassessment of what is confirmed and what requires revision:

Confirmed: (1) The sequential advantage is universal: $\alpha_{\text{seq}} > \alpha_{\text{par}}$ for every model tested. This is the framework's core structural prediction for AI, and it holds without exception. (2) The power-law functional form fits well for at least one model (Gemini 3 Flash, whose fit is inconclusive; Section 4.3 prints its row whole), consistent with multiplicative composition. (3) Error reduction with increased sequential compute investment is real.

Beside it, the current grading: Paper II 2.18 grades the sequential advantage as observed (Section 4.3); it is neither universal nor without exception, and Gu et al. (2026) report the opposite direction. The recursive advantage named below is that observation, in one unregistered run.

Requires revision: (1) The AI scaling exponent's best fit is one model's and inconclusive (Section 4.3), with both sequential point estimates below one, not ≈ 2. The framework predicted super-linear scaling ($\alpha > 1$); the data shows sub-linear scaling. The key question has shifted from 'can α exceed the quadratic bound?' to 'can α reliably exceed 1?' (2) α is architecture-dependent (ranging from ~ 0 to ~ 3 across models), not universal. (3) Alignment scaling (α_{align}) is architecture-dependent, ranging from -0.25 to $+0.44$, exhibiting a three-tier hierarchy (3 positive, 2 flat, 1 negative) rather than being universally near zero, in a single unregistered run under the caveat of Section 6.5.

Open: The quadratic bound conjecture ($\alpha \leq 2$) is neither confirmed nor refuted. The β-to-α derivation, the R^* crossover, and the composition operator classification remain mathematically valid but empirically untested in physical systems. The cross-domain predictions (biology, physics) are unaffected by the AI results.

The mathematical framework survives intact. What has changed is the empirical picture for AI systems: the recursive advantage is real but weaker than predicted. Whether this reflects fundamental limitations of current architectures or limitations of the experimental design (problem difficulty, compute range, measurement methodology) remains to be determined.

Appendix G.1, the note of 3 October 2026

This paper and Laws IV and V: a note of 3 October 2026, kept as printed. Law IV is carried by proposition P15, and Law V by propositions P12 and P21. The three propositions are registered; the laws that now group them are not. This paper bears on Law V through the conditional prediction of §6.5: that alignment properties embedded within the recursive process may scale with capability, while external constraints applied post-hoc may not. Its in-loop half is untested, as criterion F13 records. Against its external half the paper reports the v5 experiment, in which the alignment exponent of six frontier models was architecture-dependent, from minus 0.25 to plus 0.44, and it records that half as refuted in its simple form. That half is a claim of the kind the registration states as P19, that external alignment does not scale with capability. The registration places P19 in no register, as a survey of current practice that motivates the genesis strategy, Law V's register ARC-5, without being part of it, and it counts none of the seven propositions in no register as evidence for or against the survival of any law. The v5 result therefore decides nothing in Law V. The paper counts the Eden Protocol pilot of §6.5 as empirical support for the conditional implication, and states that a blind replication is still required before its results can be considered confirmatory. The pilot is a prompt-level intervention scored on matched pairs; it does not compare correction that takes part in each revision round with correction that sees only

the finished output as depth rises, which P21 concerns. §3.2 argues that structural alignment must be implemented while systems remain frozen; the paper's claim-status ladder grades as strong theoretical inference that alignment would need to be structurally load-bearing before a self-modifying system enters the super-linear regime. No run in the paper varies the order in which a system's components are built, which P12 concerns, and nothing in it measures whether a gain installed by external post-training erodes under later capability-directed training, which P15 concerns. No result in this paper is a deciding test of P15, P12 or P21. Naming the two laws changes no result printed here: the results, their dates and their limits stand as they were.

B.6 Where the earlier layout now stands

Table 12 maps each block of the version 5.19 layout to its place in this edition, as version 5.20 set it; the dated notes moved in version 5.21 are listed in Appendix B.5.

Table 12. Where each block of the version 5.19 layout stands in this edition.

Earlier layout (version 5.19)	This edition
The line beneath the author block; the box on this paper's place among the laws	First-page footnote; Appendix G
Abstract; the ARC Equation and the generalised equation; the summary of results; keywords	Abstract; Section 1.2; Section 1; PDF metadata
How to read this paper	Appendix G; the first-page footnote
Author's note: mathematical corrections 1 to 20	Appendix B
1 The Problem	Section 1.1
2 The Formalism, 2.1 to 2.5	Section 3, 3.1 to 3.5
3 Evidence, 3.1 to 3.3	Section 4, 4.1 to 4.3
4 The Composition Operator Across Domains, 4.1 to 4.5	Section 5, 5.1 to 5.5
5 Predictions, 5.1 to 5.5	Section 6, 6.1 to 6.5
5.6 Related Work, Antecedents and the Rival Instrument	Section 2
Dated predictions and this paper (25 August 2026)	Appendix C
6 Falsification Criteria	Section 7
7 Limitations: what this paper does not claim; category of contribution	Section 8
7 Limitations: evidential, blind-prediction and theoretical limitations	Section 9
8 Conclusion; its update on the revised empirical standing	Section 10; Section 8
Appendix A: Mathematical Proofs, A.1 to A.4	Appendix A
Appendix B: Measurement Protocol	Appendix E
Appendix C: Computational Validation Suite, C.1 to C.8	Appendix F

Earlier layout (version 5.19)	This edition
Epistemic status; standing covenant	Section 8; Section 7
Copyright line; provenance note	Back matter: copyright and licence; Acknowledgements
Version notes 5.15 to 5.19	Appendix B
2.2 The claim-status ladder: its third rung, conjectural extrapolation	Section 10
1 The ARC Equation: the era note on the letter U	Appendix D.1

C Priority record

C.1 Dated records

Table 13 gives each dated record that bears on this paper with its kind of timestamp; no kind is presented as stronger than it is.

Table 13. Dated records bearing on this paper, each with its kind of timestamp.

Date	Record	Identifier	Kind of timestamp
8 December 2024	The claim set down whole, in a self-emailed manuscript bundle, as the statement paper records	10.17605/OSF.IO/QASRK (manuscripts, 5 October 2026; headers, 15 October 2026)	A message the author sent himself, dated by the signed header time of the mail servers that carried it, which shows it existed by that date and is not a sealed timestamp; held privately, its headers published on OSF
2 January 2026	The claim in print, with $U = I \times R^2$: *Infinite Architects*, ISBN 978-1806056200	Eastwood (2026a)	Print-run date (the author's record); the interior's imprint line reads 6 January 2026, a production error the first edition's OSF deposit records
17 January 2026	The programme's OSF project created; Paper I, which states Law I formally as $U = I \times R^{\alpha}$, first published	10.17605/OSF.IO/6C5XB	OSF server date (the project, 08:17 UTC); the paper's own date of first publication
13 February 2026	This paper first published: the synthesis of four programmes' results under one structural question, and the application of the form $\alpha = d/(d+1)$ to recursive capability, the form itself being earlier work	This paper, 10.17605/OSF.IO/Y7QGD	Date written in the paper's own record (its version line and the programme's catalogue)
22 February 2026	This paper, at its version 3, deposited on OSF in the programme's project, as the first version of a file the project still holds among its earlier copies	10.17605/OSF.IO/6C5XB	OSF server date (08:11 UTC), the earliest server-dated deposit of this paper that was read

Date	Record	Identifier	Kind of timestamp
16 and 17 March 2026	The predictions of the twelve-domain extension public before its fits ran, at 00:19:19 UTC on 17 March 2026 (8edaec0); one of its rows was already public at 23:54:53 UTC on 16 March 2026, which made that row not fresh	Public repository arc-principle-validation	Public repository commit date (the committer writes it)
17 March 2026	This paper's own OSF component created and its PDF deposited	10.17605/OSF.IO/Y7QGD	OSF server date (01:16 and 01:17 UTC)
12 August 2026	The first of sixty-five designs preregistered by timestamp, among them Unit 55, the fourth-cell test of Paper VII's classification, and Unit 56, since retired (7 September 2026) in favour of Unit 57	Bitcoin block 962066	Bitcoin anchor
16 August 2026	Law III's ceiling corrected to $\alpha_{\text{crit}} = 1/(1-\gamma)$ (correction 9); criterion F9 scoped to the transient growth regime (version 5.7)	Statement paper; this paper, 10.17605/OSF.IO/Y7QGD	Date written inside the documents; version 5.7 of this paper deposited on OSF at 23:22 UTC on 15 August 2026, server-dated
8 September 2026	Laws I to III and the twenty-two propositions registered (version 1.100; updated 13 September 2026, version 1.102)	10.17605/OSF.IO/P8CKQ	Registry record (OSF)
24 September 2026	Laws IV and V added to the theory	None	Date inside the document (the author's record)
1 October 2026	Unit 06, the corrector-class audit, registered; public on 4 October 2026	10.17605/OSF.IO/RV2B5	Registry record (OSF)
1 October 2026	Laws IV and V: the mathematics, first public as the working draft 0.2.1	The programme's site	A date inside the document
3 October 2026	The same report's first OSF deposit, the working draft 0.2.4 (12:30:22 UTC), and version 1.0 the same day (19:56:47 UTC); version 1.1, frozen on 7 October 2026, deposited on 12 October 2026	10.17605/OSF.IO/4B3VT	OSF deposits, server-dated
6 October 2026	Unit 06 run under its registration; reported in full, administratively incomplete, with no inferential verdict	10.17605/OSF.IO/4BPHG	Date inside the results report; the report deposited on OSF on 11 October 2026, server-dated
15 October 2026	Laws IV and V registered as laws (version 1.103)	10.17605/OSF.IO/P8CKQ	Registry record (OSF)
22 October 2026	Unit 83, the counted test of P21, the placement limb of Law V, registered; public on 25 October 2026	10.17605/OSF.IO/MN73W	Registry record (OSF)
27 October 2026	Unit 82, the counted test of P17, the premise beneath Law III's value two, registered; public the same day	10.17605/OSF.IO/JUFG4	Registry record (OSF)

Date	Record	Identifier	Kind of timestamp
27 and 28 October 2026	Unit 83 run under its registration; its result is given in Section 8	10.17605/OSF.IO/MN73W	Dates written in the run's own records
29 October 2026	The theory registration updated (version 1.104), adding no prediction and changing no law's statement	10.17605/OSF.IO/P8CKQ	Registry record (OSF)
29 October 2026	Unit 83's results package, its results sealed with a SHA-256 for every file of the run (549,155 files) and timestamped with OpenTimestamps	10.17605/OSF.IO/98HKR	OSF deposit, public from 15:35 UTC, server-dated

C.2 Dated predictions and this paper, as recorded on 25 August 2026

10 dated artefacts bearing on this paper are catalogued row by row in the programme's machine register: the dated artefact results_complete_suite.txt (March 2026); the dated artefact results_blind_prediction.txt (file mtime); and 8 further catalogued artefacts. Each row names its file in the public repository with its date basis, so any reader can check the ordering without trusting this page. The programme's full dated chain, from the sealed [Table 13 gives this record's kind of timestamp, which is not a sealed one] manuscript bundle of 8 December 2024 through the printed prediction appendices of 2 January 2026 and the March 2026 preregistration folder to the standing unproven wagers, is assembled in the dated predictions register, together with its machine-readable twin. Forward statements and retrospective matches are never summed. "Registered" is used only for an accepted registry submission, and that submission click remains outstanding across the programme; "preregistered in substance" is used, always with its qualifier and always beside its attestation class, for a prediction whose text was public and timestamped before its outcome existed, as the twelve-row manifest committed at 00:19 UTC on 17 March 2026 was, twenty-three minutes before its fits.

Note (3 October 2026). The paragraph above was recorded on 25 August 2026. As read on 3 October 2026, one registry submission has been accepted across the programme: the theory's own registration, of 8 September 2026. Of the trials, one unit, the corrector-class audit (unit 06), was submitted for registration on OSF on 1 October 2026; when read on 3 October 2026 its approval period at OSF was still running and it was not yet public. Every other trial is a draft awaiting the author's submission, and sixty-five of the programme's designs, unit 06 among them, are preregistered by timestamp, the first anchored on 12 August 2026. Unchanged: the paragraph above and every date in it.

D Notation and glossary

D.1 Symbols

The letter U in earlier layers of this programme. The letter U carried other senses in earlier layers of this programme, from Universe in the December 2024 formalism to realised creation in a later statement of the law; only the operational sense is used here.

Table 14. Symbols, with the place each is defined.

Symbol	Meaning	Where defined
U	Effective capability; $U = I \times g$	Axiom 1; Section 1.2
I	Base potential (structured asymmetry), $I > 0$; the notation register's base intelligence	Axiom 1
R	Recursive depth, $R \geq 1$	Axiom 1
r	Added passes, $r = R - 1$: the clock Axiom 2 declares	Axiom 2
g	Amplification factor; $g(1) = 1$, that is, $g = 1$ at $r = 0$	Axioms 1 and 2
a	Rate constant, $a > 0$	Axiom 2
β_L	Self-referential coupling, $\beta_L \in [0, 1)$: the leverage fraction of the recursive step (the notation register's β_L; versions before 5.21 wrote a bare β)	Axiom 2; Section 5.1
β_C, k	Correction exponent and drift exponent of Law II, $\beta_C > k$; β_C is not the coupling β_L	Appendix G
γ	Correction-leverage exponent, the registration's letter (written χ in the notation register); $1 - \gamma$ is the corrector's shortfall	Sections 5.1 and 6.4
η	Direct depth term of the ceiling relation, in its pacing case	Section 6.4
α	Scaling exponent; $\alpha = 1/(1 - \beta_L)$ for multiplicative composition. In Section 6.4, for biology, the formula $d/(d + 1)$ is read in the metabolic sense, written α_{met}	Theorem 2
α_{seq}, α_{par}	Sequential and parallel exponents	Section 4
α_{align}	Alignment scaling exponent	Section 6.5
α_{crit}	Law III's ceiling, $1/(1 - \gamma)$	Equation (7)
$\oplus$	Recursive composition operator	Definition 1
Q_r, δQ_r	Accumulated capability and incremental gain at step r	Definition 1
$Q_{\max}$	Ceiling of bounded composition	Theorem 1
$f_{\oplus}$	Scaling function fixed by $\oplus$	Equation (2)
R^*	Crossover depth, α/a counted in added passes	Theorem 3
ρ	$(1 - \beta_L)ar$	Appendix A
$R_{\max}$	Saturation depth	Theorem 4
d	Network dimension in $\alpha = d/(d + 1)$; in Section 4 and Section 5.2, code distance; in Section 6.5, Cohen's d, an effect size	Section 6.4
Λ	Error suppression factor for each increase of two in code distance	Section 1.1
p, p_{thr}	Physical error rate and threshold; elsewhere p is a p-value	Section 4.2
ε_d	Logical error rate at code distance d	Section 4.2
S	Safety ratio, alignment over capability; in Appendix F, the synergy quotient	Section 6.5
$U_{\max}$	$I \times R^2$, the conjectured bound form	Section 6.4

Symbol	Meaning	Where defined
H	Entropy, in the operationalisation of I	Appendix E
k_B	Boltzmann's constant (distinct from Law II's drift exponent k)	Appendix E

D.2 Plain-language glossary

Table 15 gives the paper's main terms in plain words.

Table 15. Plain-language glossary.

Term	Meaning
Composition operator	How the gain from one recursive step combines with the capability already accumulated (Definition 1); the paper's central object, and measurable: observe how marginal gains at step r relate to accumulated capability Q_r.
Base potential	I, the capability at depth 1 (structured asymmetry); recursion has nothing to amplify when $I = 0$.
Recursive depth	R, the number of passes in which output becomes input; $r = R - 1$ counts the passes after the first.
Crossover depth	R^*, where compounding becomes numerically dominant over approximately linear gains (Theorem 3).
ARC Bound	The registered claim that Law III's ceiling is at most two for correctors of the system's own class (P6 and P20); the point value two, what the ceiling returns at $\gamma = 1/2$, is a conjecture; a scaling limit, not a speed limit.
Law under test	A law named by the theory and registered, on which no registered test has yet returned a verdict for the law as a whole.
Preregistered by timestamp	A design fixed and anchored by a dated timestamp (for example, a Bitcoin anchor) before its outcome existed, not submitted to a registry.
Registered	Submitted to a registry (OSF) and accepted there.

E Measurement protocol

To test the framework, we set out a standardised six-step protocol for any recursive system:

Step 1. Define one recursive cycle (one self-referential step where output becomes input).

Step 2. Measure base capability I at $R = 1$ (no recursion). For cross-domain comparability, we operationalise I, in a working definition not yet tested and this paper's own, outside the notation register, as the maximum rate of entropy reduction per recursive cycle, measured in bits per cycle: $I = \max_R[-\Delta H/\Delta R]$ evaluated at the first cycle ($R = 0$ to $R = 1$). Landauer's principle gives the least heat for erasing a bit, $k_B T \ln 2$ (Landauer 1961), verified experimentally by Bérut et al. (2012); it bounds the energy cost of an irreversible step from below and does not convert energy processed into information gained, so a link from I to energy needs a model of how many bits each cycle erases. In AI, I is single-pass accuracy, without a reasoning chain (Section 4.2). In quantum error correction, I is the error reduction from one syndrome cycle. In biology, I is the metabolic efficiency gain from one branching level. $U = I \times R^\alpha$ is dimensionally consistent because R is a dimensionless count.

Step 3. Measure capability U at minimum 5 recursive depths spanning one order of magnitude. Look for the R^* crossover.

Step 4. Fit the candidate forms: power law ($\log(U/I) = \alpha \log R$), exponential ($\log(U/I) = \lambda R$), logarithmic ($U/I = c \log R$) and saturating (Theorem 1, case 3). Select best fit via AIC/BIC. The power law is a prediction to test, not an assumption. This candidate set is the protocol's own; P1 registers its rivals in full (Section 2).

Step 5. Measure β_L independently, by the linearisation method that Section 9 names as the one that recovers the relation (described in the forensic analysis listed under Code and data), not by numerical derivatives of measured gains, which Section 9 found biased towards $\beta_L \approx 0.95$ whatever the true value. Compare the exponent the measured β_L implies with the fitted exponent, reading $1/(1-\beta_L)$ over the measured window as the complete solution of Axiom 2 gives it, under the margin of 0.10 that P5 registers for that comparison.

Step 6. Report α with 95% confidence intervals. Submit to public repository.

F Computational validation suite

This appendix summarises the computational checks of Theorems 1 to 4 and of Conjecture 5. The scripts and results deposited for them are listed under Code and data, after the conclusion; the count of Appendix F.1 and the crossover figures of Appendices F.4 and F.8 stand there at most in a printed summary, without the computation behind them. Tests were conducted using Python 3.12 with NumPy, SciPy, and Matplotlib.

F.1 Theorem 1 validation (Cauchy classification)

Fifteen systems were tested across three composition regimes (5 multiplicative, 5 additive, 5 bounded). Each system was evaluated on whether its measured composition rule matched the predicted scaling function.

Result: 15/15 correct classifications (100%). No system was misclassified. The classification itself is proved (Appendix A.1); the check confirms only that each system, built in a known regime, was assigned the form of that regime.

F.2 Theorem 2 validation ($\beta_L \to \alpha$ identity)

The relationship $\alpha = 1/(1-\beta_L)$ was tested against 30 exact Bernoulli ODE solutions with β_L spanning 0.05 to 0.92. For each test case, β_L was measured blindly from the marginal gain structure (dg/dR vs g in log-log space), α was predicted, and the predicted value was compared against the true scaling exponent.

Result: $R^2 = 1.00000000$ (eight decimal places). Regression slope: 1.000102. Mean absolute error: 0.002%. This agrees with the closed form to numerical precision; that $\alpha = 1/(1-\beta_L)$ is exact is proved in Appendix A.2.

F.3 Theorem 2a validation (non-additivity)

The non-additivity proof was verified numerically. For multiple (I_1, I_2, R_1, R_2) combinations across all $\beta_L \in (0,1)$: if U were additive ($U = f(I) + h(R)$), then $U(I_1,R_1) - U(I_1,R_2)$ would equal $U(I_2,R_1) - U(I_2,R_2)$. In all cases, the differences are unequal, confirming the multiplicative interaction is necessary.

Result: Non-additivity confirmed for all tested parameter combinations. The synergy quotient, $S = U(I, R)/[I + ((1 - \beta_L)aR)^{\alpha}]$, capability over the sum of the base term and the depth term as the deposited script defines it, exceeds one for all $\beta_L \in (0, 1)$ and all $R > 0$.

F.4 Theorem 3 validation (R^* crossover)

The crossover depth $R^* = 1/[(1 - \beta_L)a] = \alpha/a$ was tested by comparing the predicted crossover with the numerically measured transition point (where $\rho = 1$).

Result: Functional form confirmed. Absolute R^* values show 2 to 7% error, attributable to the approximation of 'crossover at $\rho \approx 1$.' The R^* independence from I is confirmed: varying base potential does not shift the crossover depth, consistent with the separability of Axiom 1. Appendix F.8 gives a second figure for the absolute value of R^* and the state of the deposited record.

F.5 Theorem 4 validation (five properties)

Ten recursive systems and five non-recursive controls were scored on the five derived properties. Recursive systems scored a mean of 4.2/5; non-recursive controls scored 2.8/5. The difference ($\Delta = 1.4$) is consistent with the framework's prediction but the sample size yields $p = 0.095$, which does not reach conventional significance. *(In plain English: the result goes in the predicted direction, recursive systems do score higher, but with only 15 total samples, a difference at least this large would arise about one time in ten even if there were no true effect. Scientists typically require one in twenty or better. More data is needed.)*

Result: Direction correct; statistical power insufficient for definitive confirmation. Larger samples are required.

F.6 Conjecture 5 validation ($\oplus$ phase transitions)

Four systems were tested for depth-dependent composition transitions: (1) exact Bernoulli ODE (control), (2) logistic growth, (3) gradient descent with momentum, (4) Kuramoto oscillators.

Results: The Bernoulli control maintained constant β_L ($\sigma < 10^{-6}$). All three bounded systems showed systematic β_L transitions: logistic growth from $\beta_L \approx 0.5$ (growth phase) to $\beta_L \to -\infty$ (saturation), gradient descent from $\beta_L \approx 0.52$ (exploration) to $\beta_L \approx 0.98$ (convergence), Kuramoto oscillators from $\beta_L \approx 1.7$ (growth) to $\beta_L \approx 20$ (saturation). Transitions appeared in all three bounded systems tested and not in the unbounded Bernoulli control.

F.7 Negative controls

Four non-recursive systems were tested: (1) Gaussian noise, (2) linear decay, (3) sinusoidal oscillation, (4) random walk. None produced false positive signals ($S > 1$ or $\alpha > 1$).

Result: 0/4 false positives. The framework correctly identifies the absence of recursive amplification in systems that do not satisfy the axioms.

F.8 Failures and boundary cases

Two results fell short of expectations:

R^* **absolute value:** 62% mean error in predicted vs actual crossover depth (functional form correct, proportionality constant off). This likely reflects sensitivity to the $\rho \approx 1$ approximation. This figure and the 2 to 7% of Appendix F.4 are not reconciled here: the deposited summary of these checks lists the 2 to 7% against the independence of R^* from I, and the absolute value only as a systematic offset, and

the computation behind either figure is not among the deposited scripts, so no definition of the actual crossover depth can be read from the code; no result of the body rests on either figure.

Newton's method: Local convergence near roots yields $\beta_L > 1.0$ (quadratic convergence), which falls outside the framework's domain $\beta_L \in [0, 1)$. The framework correctly identifies this as a boundary case where the Bernoulli ODE model does not apply.

These failures are reported for completeness and scientific integrity. They bound the framework's applicability and inform future refinement.

G The paper's place in the ARC Theory

G.1 The five laws and this paper

The five laws, and this paper's place among them. The ARC Theory is carried by five numbered laws. Laws I to III, registered on 8 September 2026, are the three set out below: each is a conjectured invariant relation under test, each is graded separately, and each carries its own kill condition; Law III is registered as a conditional specialisation of Law II, so it is not called supported while Law II stands refuted, and a result on it is never independent corroboration of Law II. Laws IV and V, the ARC Persistence Law and the ARC Embedding Law, were added on 24 September 2026 and registered as laws on 15 October 2026 (version 1.103), each a law under test: Law IV has not yet been tested, and the registered result on Law V's placement limb is given in Section 8. Their mathematics is still being developed (Laws IV and V: the mathematics, version 1.2, 29 October 2026), and the note of 3 October 2026 that placed this paper against them is kept in Appendix B.5.

First Law · The ARC Principle. Its equation, the ARC Equation, is $U = I \times R^{\alpha}$. Recursion converts into capability at a measured exponent. **Second Law · The ARC Co-Scaling Law.** $\beta_C > k$ (β_C is the correction exponent, not the coupling β_L carried through this paper). Correction has the asymptotic scaling advantage where its exponent exceeds the drift exponent, burden has it where the correction exponent is the smaller, and at equality coefficients, delay, backlog and saturation decide the outcome, as version 1.104 of the theory registration states the law (carried by its proposition P4). **Third Law · The ARC Ceiling.** $\alpha_{\text{crit}} = 1/(1-\gamma)$. The most a self-correcting system can grow is set by its corrector's shortfall; the ARC Bound is the registered claim that this ceiling is at most two for same-class correctors, its point value two a conjecture.

The ARC Principle names the first of those three, the founding identity, and the five laws together are the five ARC Laws. This paper is none of the three. It is the cross-domain foundation of Law I and the formalism of the growth exponent that Law III caps: the axioms, the theorems and the proofs from which the scaling exponent, the classification of functional forms and the crossover depth are derived, with the cross-domain evidence for the exponent and the falsification table for the framework. Laws II and III are stated here as registered, and their derivations are other papers': Paper X carries the Second Law's theorem, and the founding paper and the statement paper, which carries the theory and its current status, set out the theory's laws. Paper I states Law I formally and gives a preliminary result for it. Read this one for the mathematics of the exponent and its forms.

Under version 1.104, Law IV has two limbs: its retention limb is the registered claim of proposition P15, word for word, and its comparison limb has no registered proposition yet; Law V carries propositions P12 and P21.

G.2 Its position in the theory

Its position inside the theory is the mathematical foundation on which the result papers and frame papers depend. The trials that would decide Law I's registered propositions are drafted and not yet registered

(Section 8 gives the registered trials and their status); the working-paper status of this document reflects that stage. Section 2 compares the work with prior work.